



MOIZ DAWOODI
O/A LEVEL PHYSICS

O' LEVEL PHYSICS

Electromagnetic Effects

4.5

STUDENT NAME



0315-0805775

PHYSICS WITH
MOIZ DAWOODI

SYLLABUS OUTLINE:

4.5 Electromagnetic effects

4.5.1 Electromagnetic induction

- 1 Describe an experiment to demonstrate electromagnetic induction
- 2 State that the magnitude of an induced e.m.f. is affected by:
 - (a) the rate of change of the magnetic field or the rate of cutting of magnetic field lines
 - (b) the number of turns in a coil
- 3 State and use the fact that the effect of the current produced by an induced e.m.f. is to oppose the change producing it (Lenz's law) and describe how this law may be demonstrated

4.5.2 The a.c. generator

- 1 Describe a simple form of a.c. generator (rotating coil or rotating magnet) and the use of slip rings and brushes where needed
- 2 Sketch and interpret graphs of e.m.f. against time for simple a.c. generators and relate the position of the generator coil to the peaks, troughs and zeros of the e.m.f.

4.5.3 Magnetic effect of a current

- 1 Describe the pattern and direction of the magnetic field due to currents in straight wires and in solenoids and state the effect on the magnetic field of changing the magnitude and direction of the current
- 2 Describe how the magnetic effect of a current is used in relays and loudspeakers and give examples of their application

4.5.4 Forces on a current-carrying conductor

- 1 Describe an experiment to show that a force acts on a current-carrying conductor in a magnetic field, including the effect of reversing:
 - (a) the current
 - (b) the direction of the field
- 2 Recall and use the relative directions of force, magnetic field and current
- 3 Describe the magnetic field patterns between currents in parallel conductors and relate these to the forces on the conductors (excluding the Earth's field)

4.5.5 The d.c. motor

- 1 Know that a current-carrying coil in a magnetic field may experience a turning effect and that the turning effect is increased by increasing:
 - (a) the number of turns on the coil
 - (b) the current
 - (c) the strength of the magnetic field
- 2 Describe the operation of an electric motor, including the action of a split-ring commutator and brushes

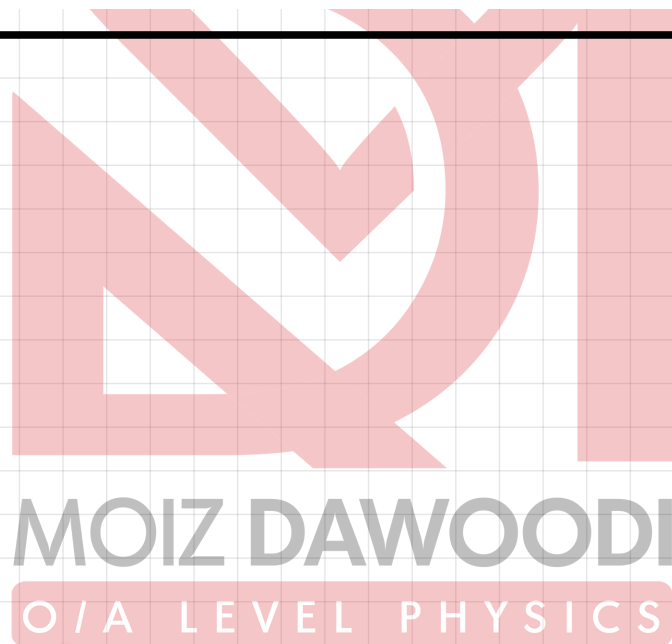
4.5.6 The transformer

- 1 Describe the structure and explain the principle of operation of a simple iron-cored transformer
- 2 Use the terms primary, secondary, step-up and step-down
- 3 Recall and use the equation

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

where P and S refer to primary and secondary

- 4 State the advantages of high-voltage transmission and explain why power losses in cables are smaller when the voltage is greater



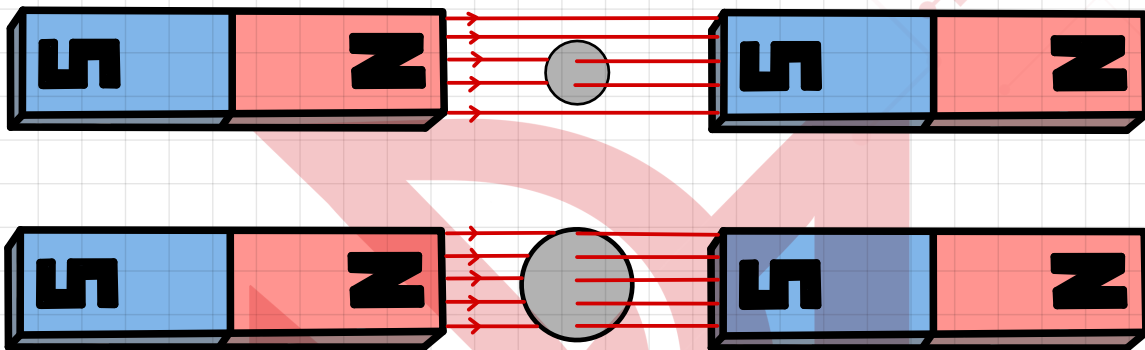
MAGNETIC FLUX:

"No. of magnetic field lines linking the coil."

FACTORS EFFECTING FLUX

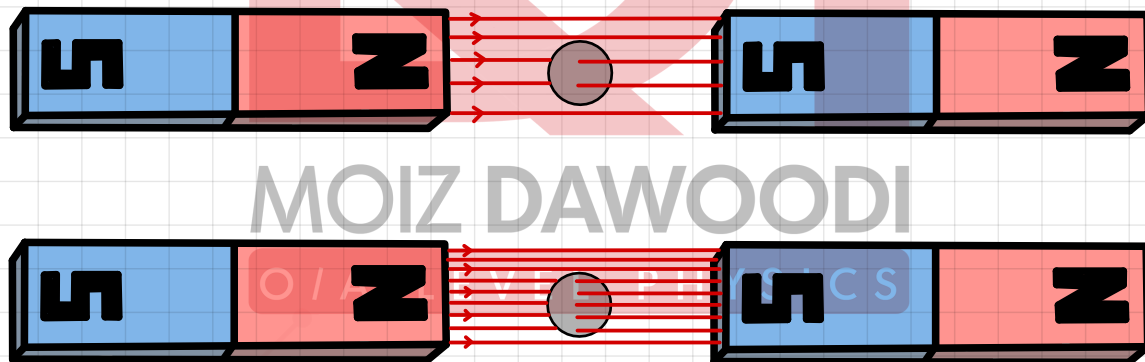
SURFACE AREA:

Magnetic Flux $\propto$ Linked Area



MAGNETIC FIELD STRENGTH:

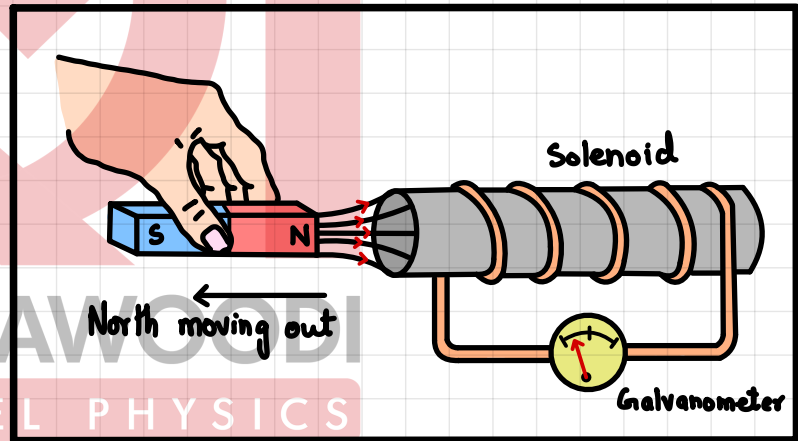
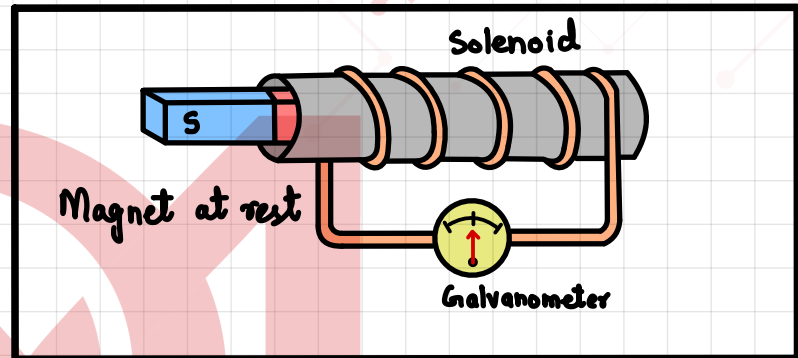
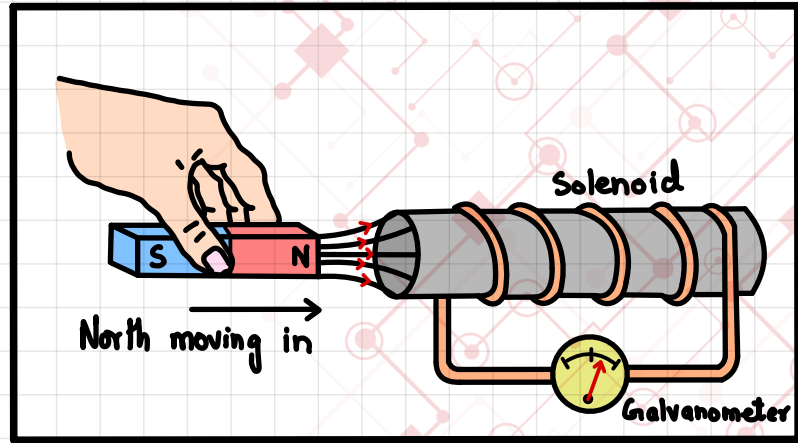
Magnetic Flux $\propto$ Magnetic Field Strength



Flux can also be changed by moving the magnet towards or away from the coil.

SOLENOID EXPERIMENT:

- When north of the magnet is pushed inside the solenoid, solenoid experiences induced current & needle moves towards right.
- When magnet is at rest either inside or outside the solenoid, no current flow in the solenoid & needle remains at centered zero.
- When north of the magnet is pulled outside the solenoid, solenoid experiences induced current & needle moves towards left.



This is the beginning concept of an AC Generator.

LAWs OF ELECTROMAGNETISM:

FARADAY'S LAW:

"Induced emf is directly proportional to the change in flux per unit time."

Flux can be increased by:

- Using stronger magnet.
- Using larger linking area.
- Increasing the speed of magnet.

This law give the size/magnitude of induced emf.

LENZ'S LAW:

جس سے فائو اُٹھتا ہے
اُسی کو نقصان پہنچاتا ہے

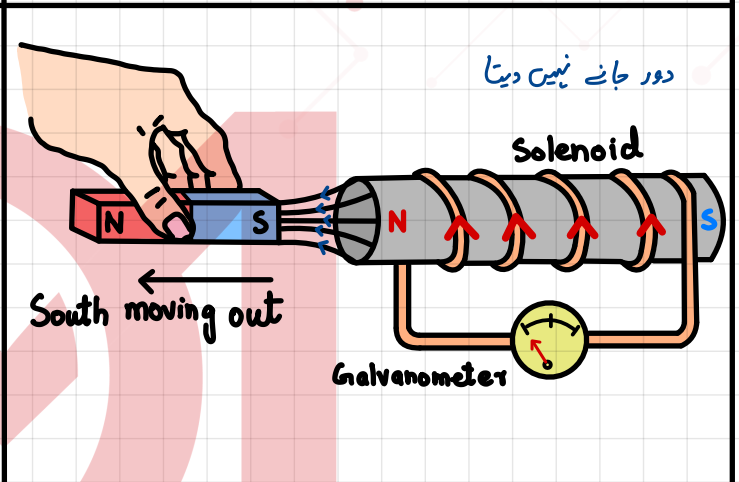
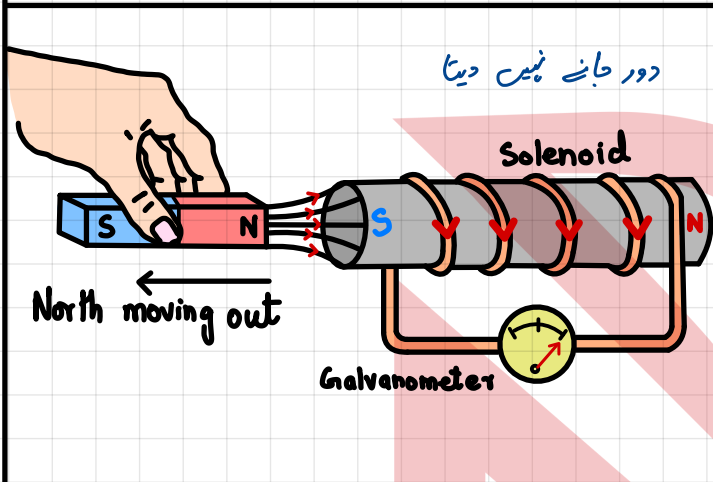
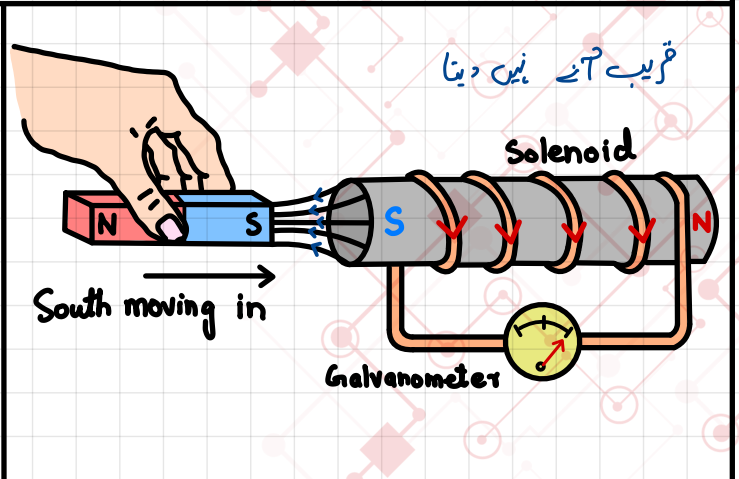
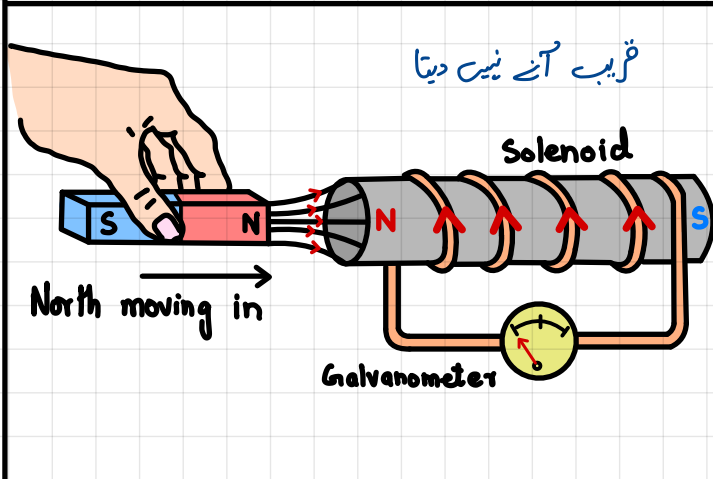
جس تھاک میں کھاتا ہے
اُسی میں چھید کرتا ہے

"Induced current always flow in a direction which opposes the cause of its induction."

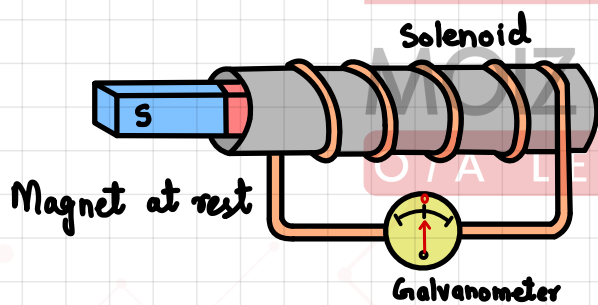
This law gives the direction of emf.

قریب آنے نہیں دیتا
دور جانے نہیں دیتا

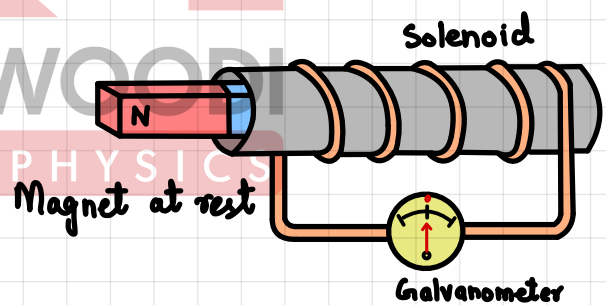
SITUATIONS



As magnet is at rest, no change in flux occurs hence no emf is induced so current doesn't flow & solenoid doesn't magnetise.



As magnet is at rest, no change in flux occurs hence no emf is induced so current doesn't flow & solenoid doesn't magnetise.



AC GENERATOR:

"A device which converts mechanical energy into electrical energy in the form of alternating current is called an AC Generator."
It consists of:

● SLIP RINGS:

They make sure coil doesn't get tangled & they allow direction of current to be reversed.

● CARBON BRUSHES:

Graphite/carbon brushes are used to make good electrical contact & it provides lubrication.

● COIL:

Coil made from an insulated copper wire.

● FIELD MAGNET:

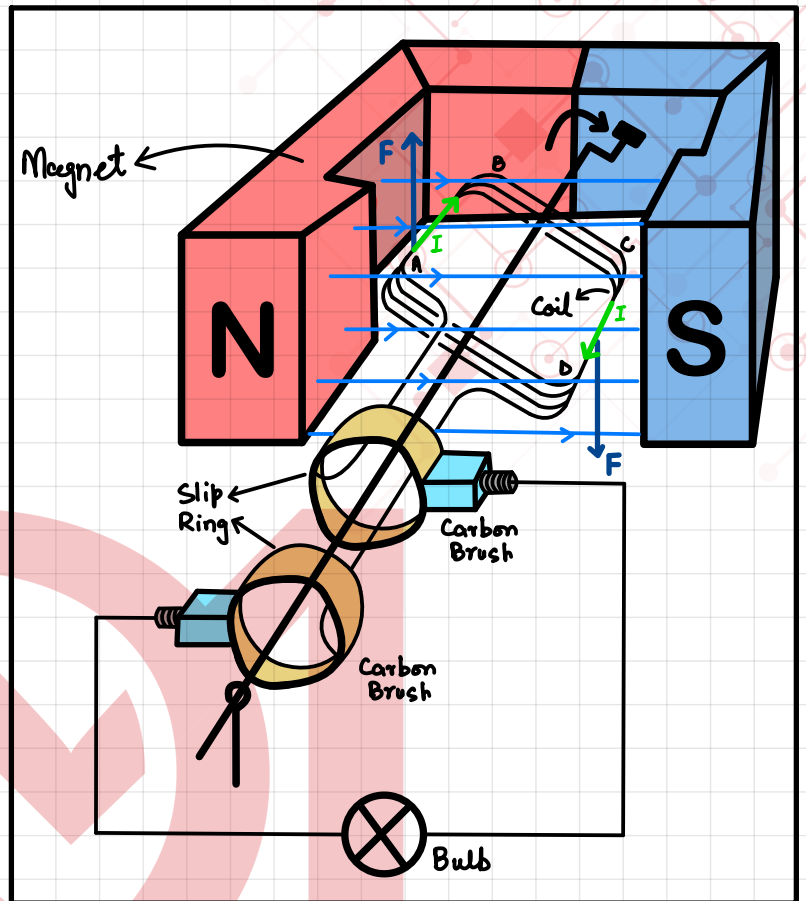
A strong permanent magnet.

FLEMING'S RIGHT HAND RULE:

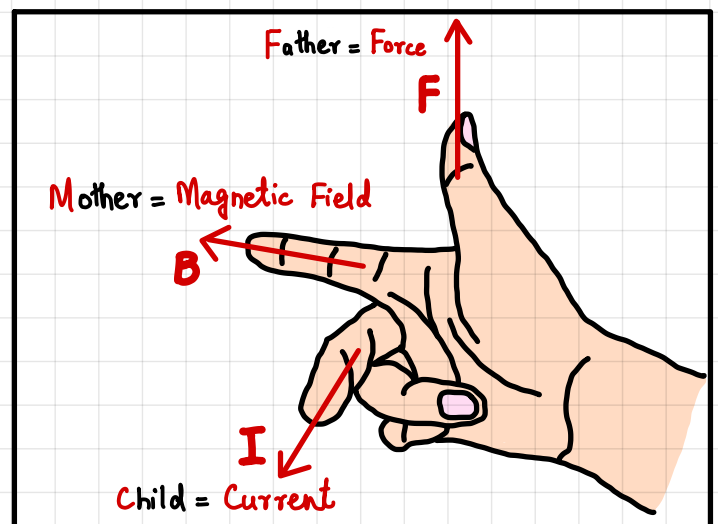
Direction of current can be found using Fleming's Right Hand Rule.

SEQUENCE.

- Align your first finger into the direction of field.

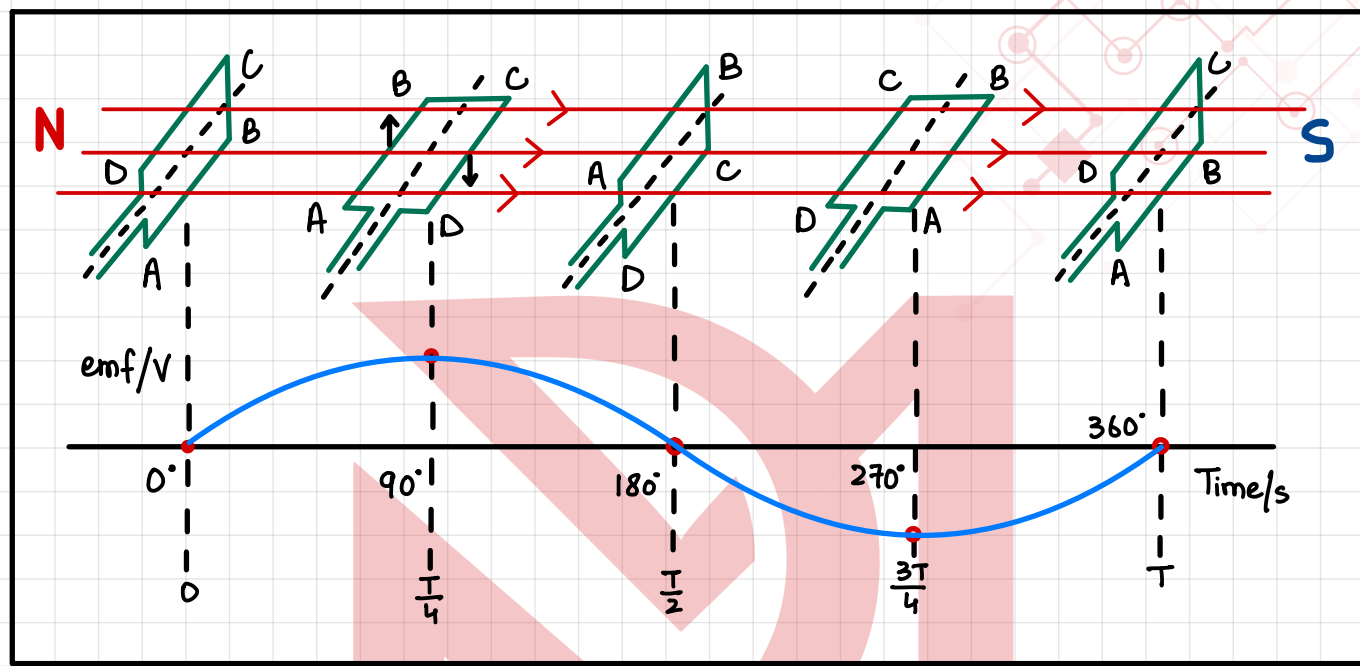


Fleming's Right Hand Rule is used when current is unknown & force is known.



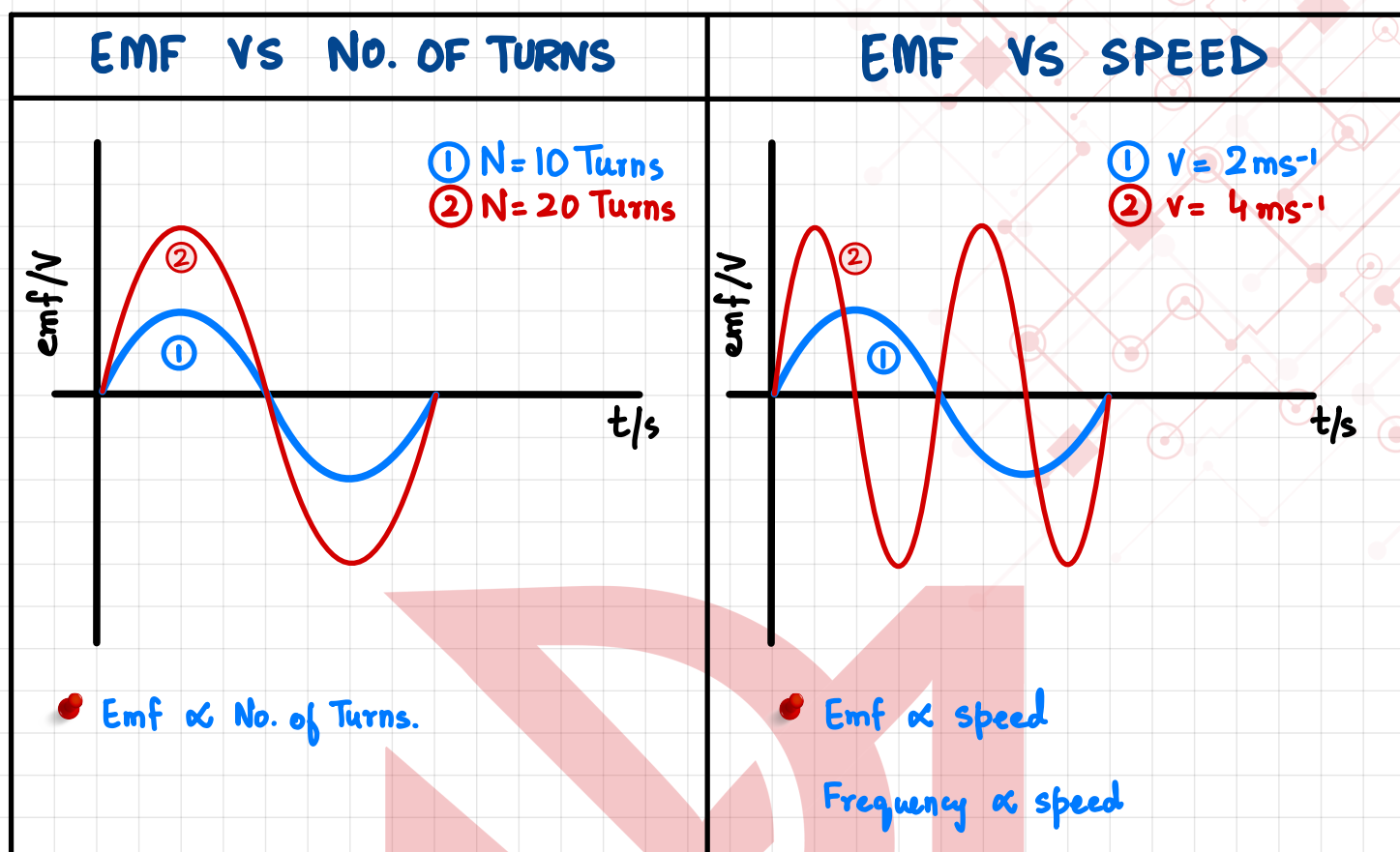
- Without moving your first finger, align your thumb in the direction of force.
- Finally your second finger indicate direction of current.

WORKING OF AN AC GENERATOR:



- 0°** : Coil is at rest, no change in flux, no emf.
- 90°** : Coil rotates quarter, change in flux goes minimum to maximum so induced emf also goes minimum to maximum.
- 180°** : Coil rotates half, change in flux goes maximum to minimum so induced emf also goes maximum to minimum.
- 270°** : Coil rotates three-quarter, change in flux goes minimum to maximum so induced emf also goes minimum to maximum. But this time direction of coil is opposite so emf is negative maximum.
- 360°** : Coil rotates full, change in flux goes maximum to minimum so induced emf also goes maximum to minimum.

GRAPHS OF AN AC GENERATOR:



TYPES OF AC GENERATOR:

- **MAGNETO.** An AC generator in which coil rotates while magnet at rest.
- **DYNAMO:** An AC generator in which magnet rotates while coil at rest.

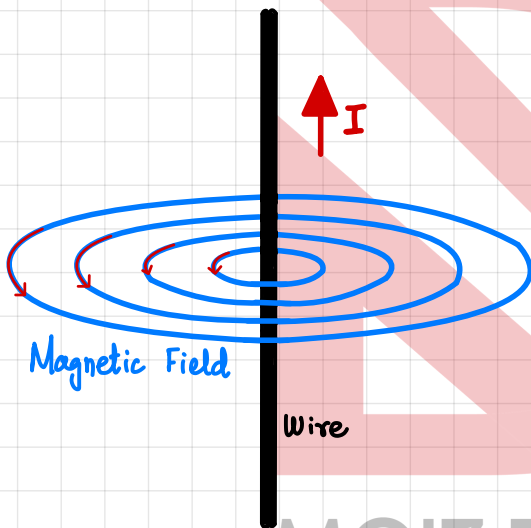
O/A LEVEL PHYSICS

ELECTROMAGNETISM

"Materials which generate magnetism when electric current flows through them are called electromagnets & the phenomenon is called electromagnetism.

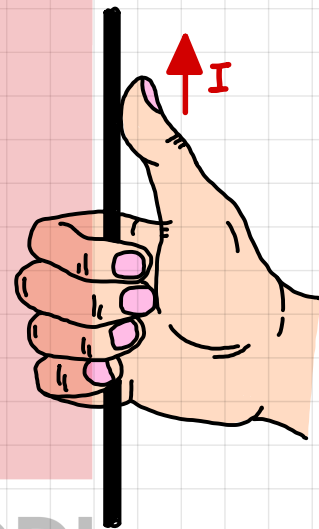
STRAIGHT CURRENT CARRYING WIRE:

STRAIGHT CURRENT CARRYING WIRE AS AN ELECTROMAGNET



When straight wire carries current, it generates magnetic field in the shape of concentric circles.

MAXWELL'S RIGHT HAND RULE

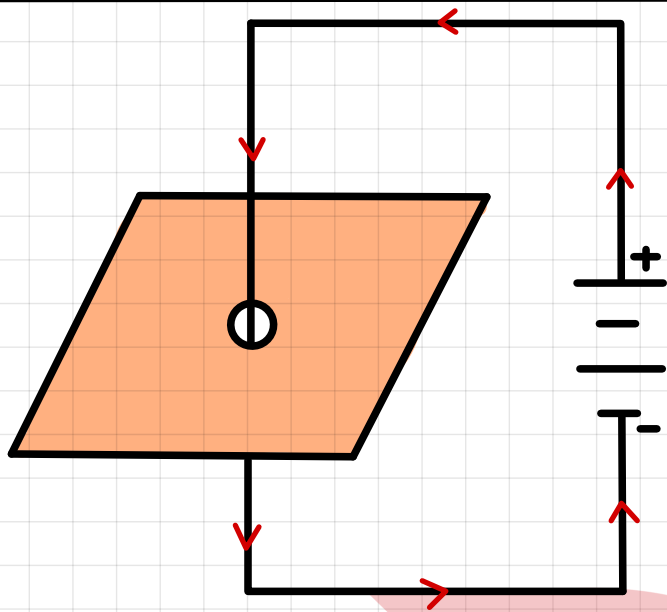


If you point the thumb of your right hand in the direction of current, the curled fingers of your right hand indicates the direction of magnetic field

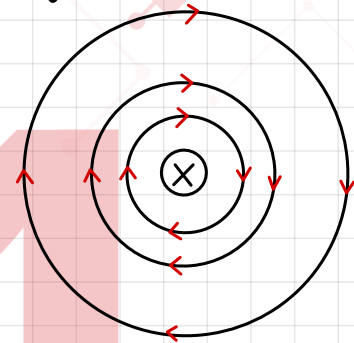
- When size of current is increased, magnetic field strength also increases.
- When direction of current is reversed, direction of magnetic field is also reversed.



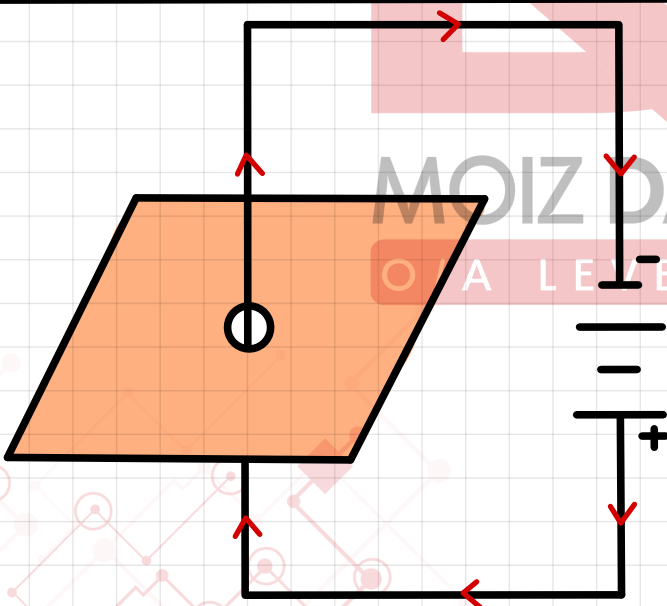
MAGNETIC FIELD OF AN INWARD CURRENT



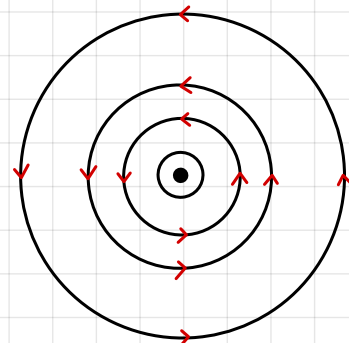
- Concentric circle.
- Field circles are closer near the wire as magnetic field is stronger.
- Inward current is represented by 'x'.
- Clockwise field.



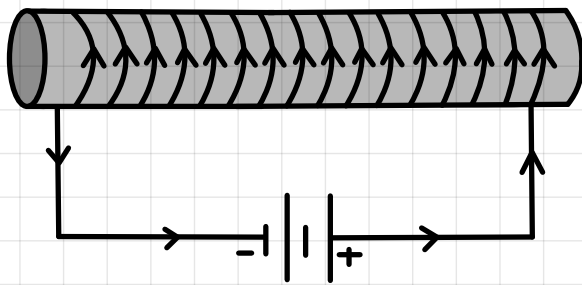
MAGNETIC FIELD OF AN OUTWARD CURRENT



- Concentric circle.
- Field circles are closer near the wire as magnetic field is stronger.
- Outward current is represented by 'o'.
- Anticlockwise field.

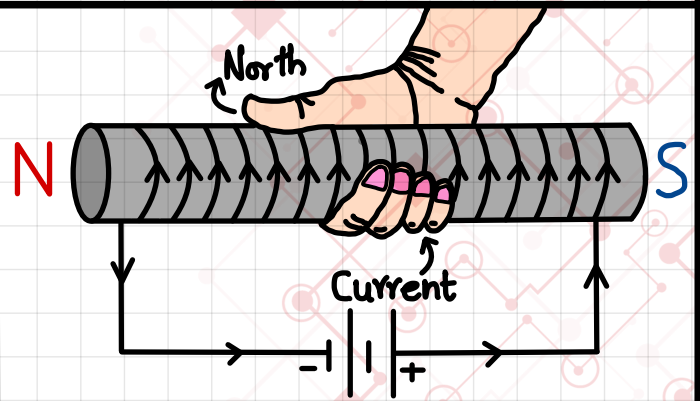


SOLENOID AS AN ELECTROMAGNET



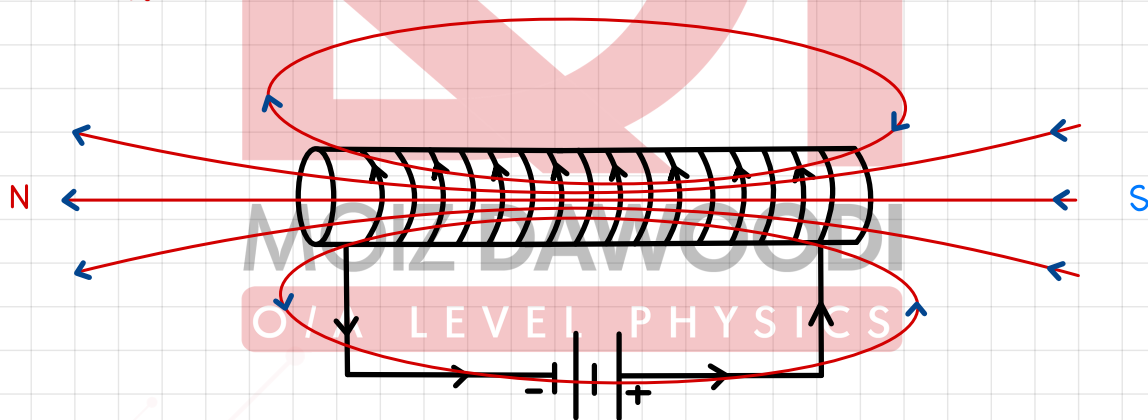
Solenoid is an electromagnet which is made by coiling a wire in the shape of spring such that each turn of coil is equidistant.

RIGHT HAND GRIP RULE



If you curl the fingers of your right hand in the direction of current. Your thumb indicates the north pole of solenoid.

DRAW MAGNETIC FIELD PATTERN OF A SOLENOID:



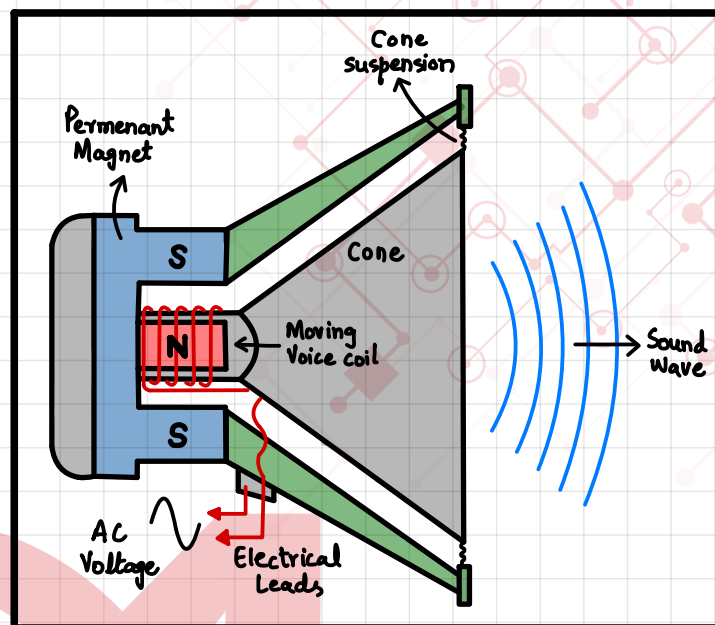
FACTORS EFFECTING MAGNETIC FIELD STRENGTH OF SOLENOID

No. of turns of coil	Magnetic Field Strength $\propto$ No. of turns.
Amount of Current	Magnetic Field Strength $\propto$ Size of current.
Use of an Iron Core	Magnetic Field Strength increases with iron core.

APPLICATIONS OF A SOLENOID:

LOUDSPEAKER:

When alternating current flows through solenoid. During the positive half cycle, solenoid generates north at the face of north of permanent magnet. This causes repulsion & coil pushes cone outward. This creates compression in air.



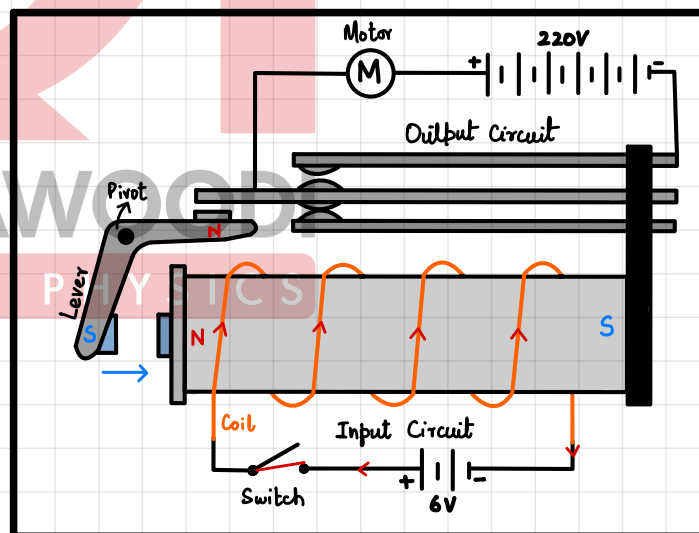
During the negative half cycle, solenoid generates south at the face of north of permanent magnet. This causes attraction & coil pulls cone inward. This creates rarefaction in air.

MAGNETIC RELAY:

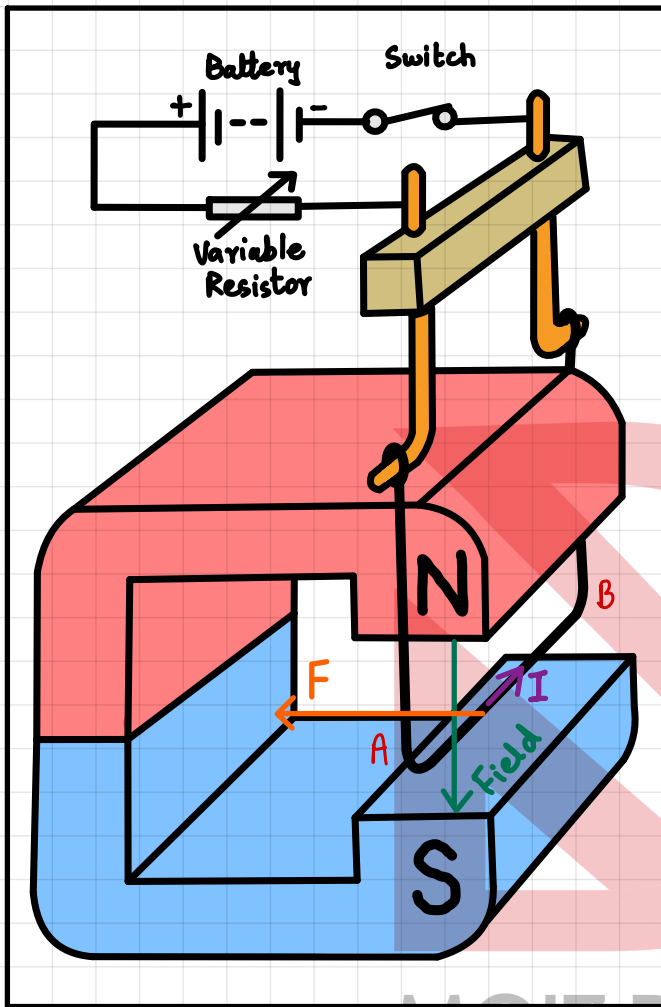
Magnetic relay is a device which switches ON high current circuit using low current circuit.

When input circuit is switched ON, solenoid magnetises & attracts iron lever. Iron lever turns & the upper part of lever pushes contacts with each other & output circuit is complete.

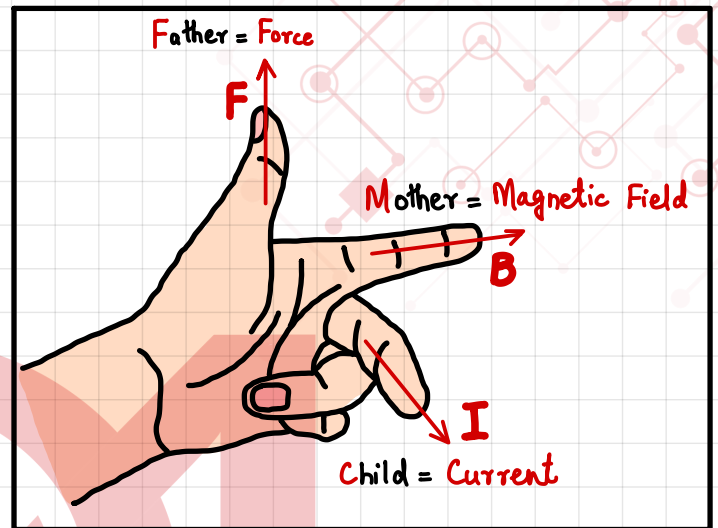
To switch OFF output circuit, input circuit is switched OFF.



FORCE ON A CURRENT CARRYING WIRE:



FLEMING'S LEFT HAND RULE:



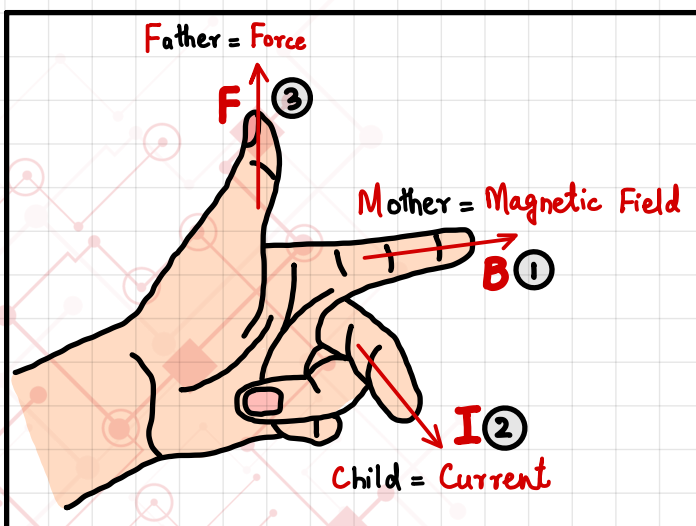
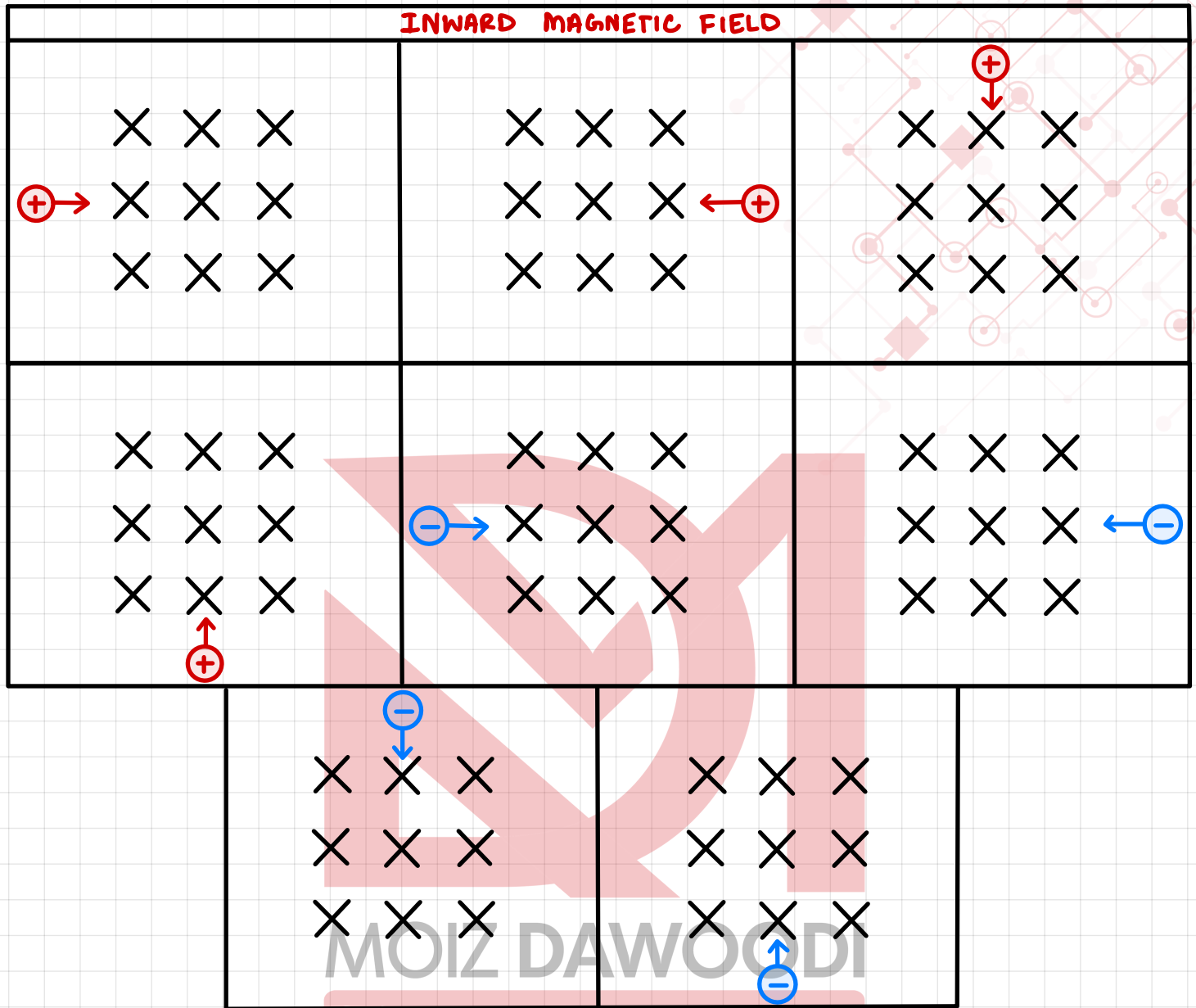
SEQUENCE.

- Align your first finger into the direction of field.
- Without moving your first finger, align your second finger in the direction of current.
- Finally your thumb will indicate the force generated.

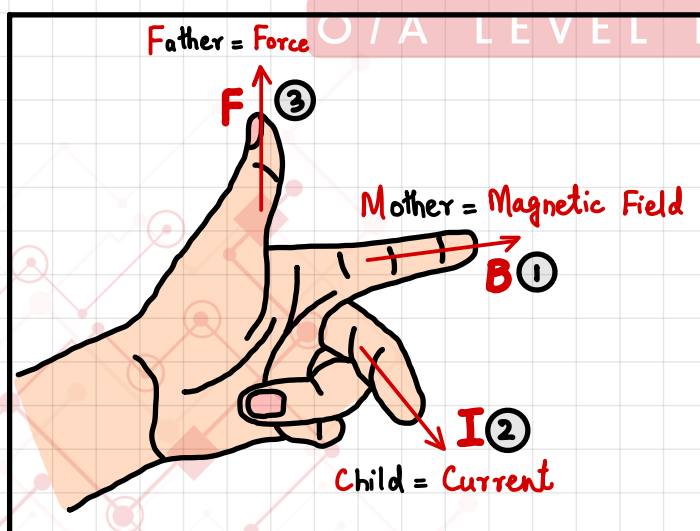
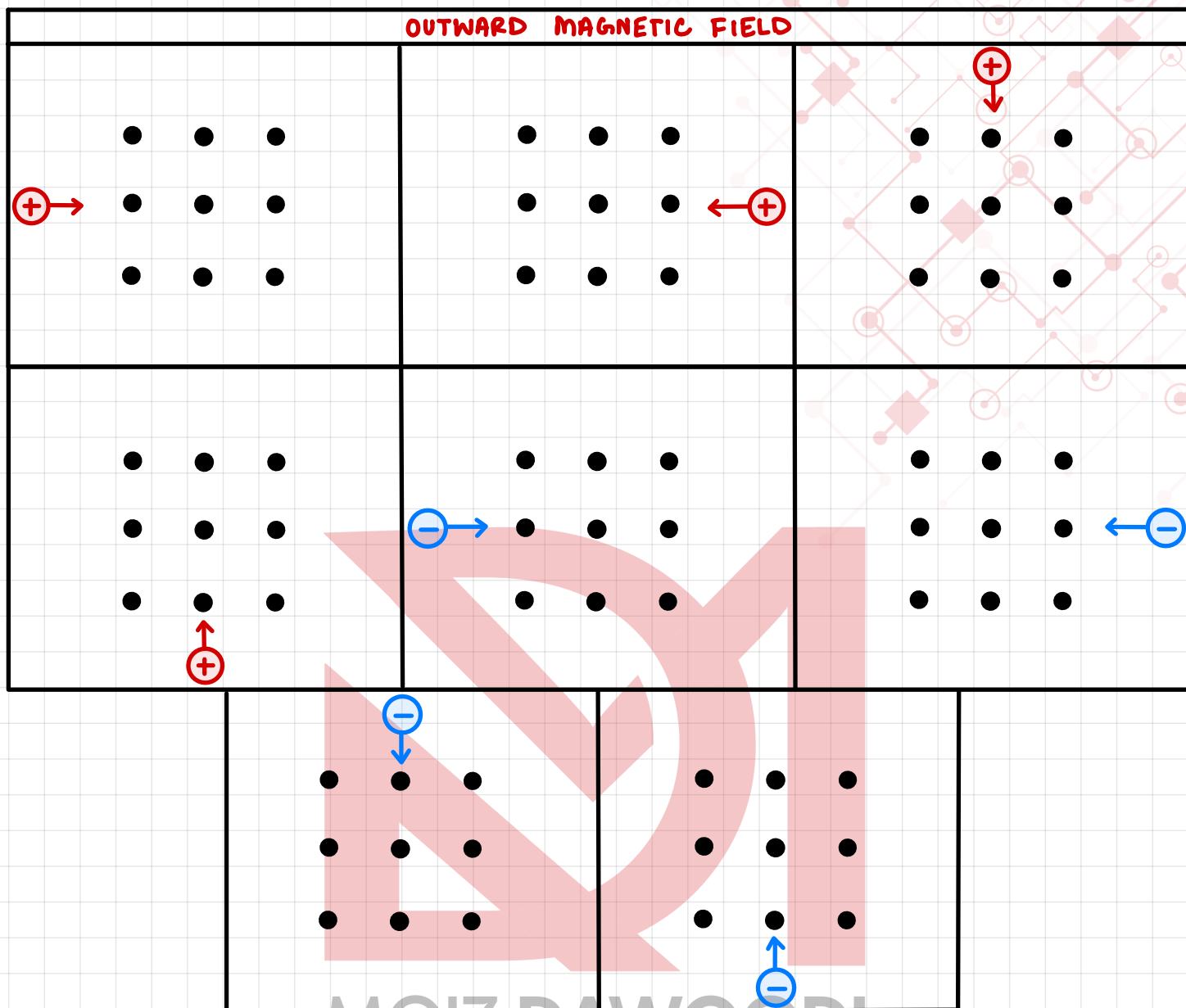
Whenever current carrying wire is placed in a magnetic field, it experiences a force. In order to find the direction of force, we use Fleming's Left Hand Rule.

- If we reverse direction of current, force will be reversed.
- If we reverse poles of magnet, force will be reversed.
- If we reverse both, force will be same.

FORCE ON MOVING CHARGE IN MAGNETIC FIELD:

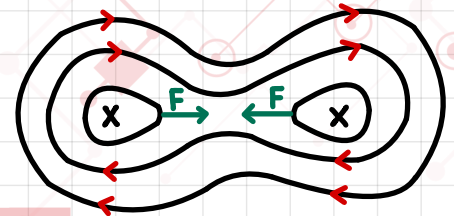
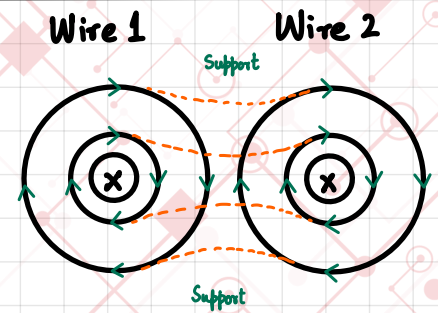
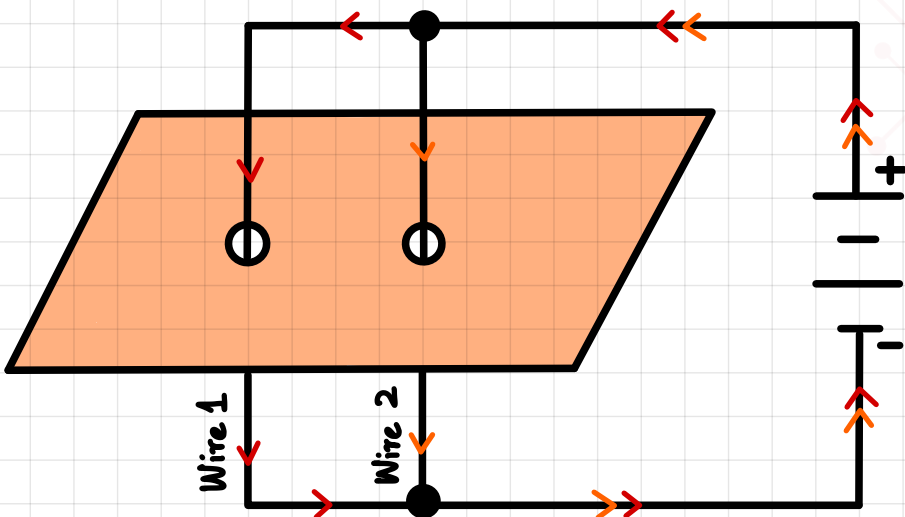


Since Fleming's Left Hand Rule is designed for Positive charge, for negative charge, reverse the direction of charge.



Since Fleming's Left Hand Rule is designed for Positive charge, for negative charge, reverse the direction of charge.

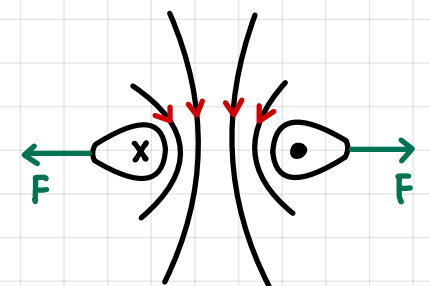
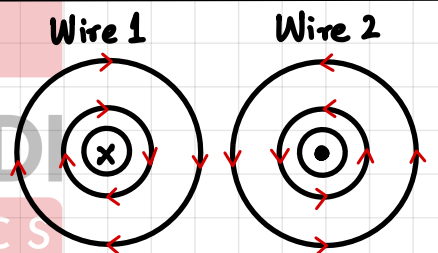
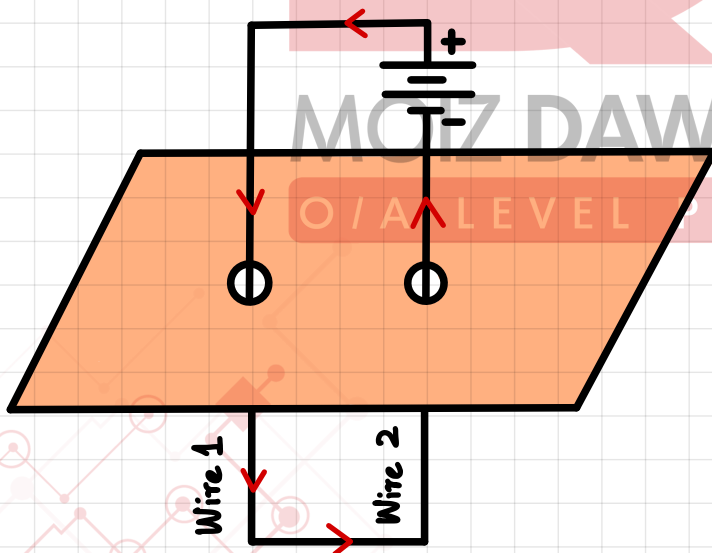
MAGNETIC FIELD OF LIKE CURRENTS



- Since the patterns of both wires are meeting, its attraction.
- Forces of wires will be towards each other.

• Like currents attract.

MAGNETIC FIELD OF UNLIKE CURRENTS

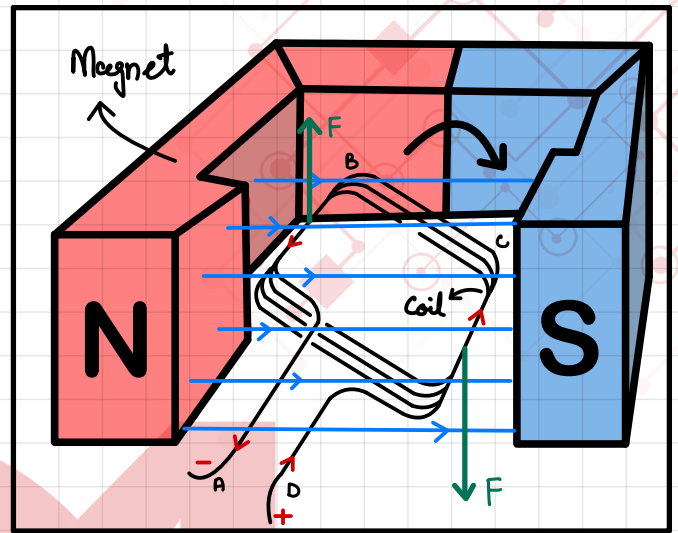


- Since the patterns of both wires are not meeting, its repulsion.
- Forces of wires will be against each other.

• Unlike currents repel.

FORCE ON A CURRENT CARRYING COIL:

When a current carrying coil is placed in a magnetic field, one side of coil experiences upward force while other side experiences downward force & coil rotates.



- If no. of turns are increased, speed of motion of coil also increases.
- If current is increased, speed of motion of coil also increases.
- If strength of magnet is increased, speed of motion of coil also increases.
- Reversing either current or magnetic poles result in reversing the rotation of coil.

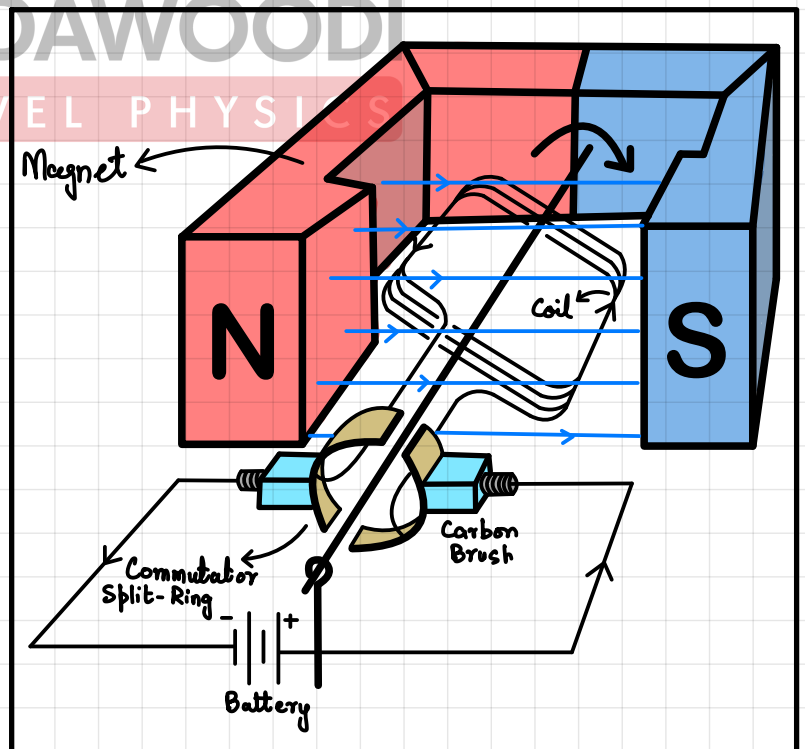
DC MOTOR:

"A device which converts electrical energy into mechanical energy."

It consists of :

SPLIT RING COMMUTATOR:

It makes sure coil doesn't get tangled & it keeps the current in one direction only.



- **CARBON BRUSHES:**

Graphite/carbon brushes are used to make good electrical contact & it provides lubrication.

- **COIL:**

Coil made from an insulated copper wire.

- **FIELD MAGNET:**

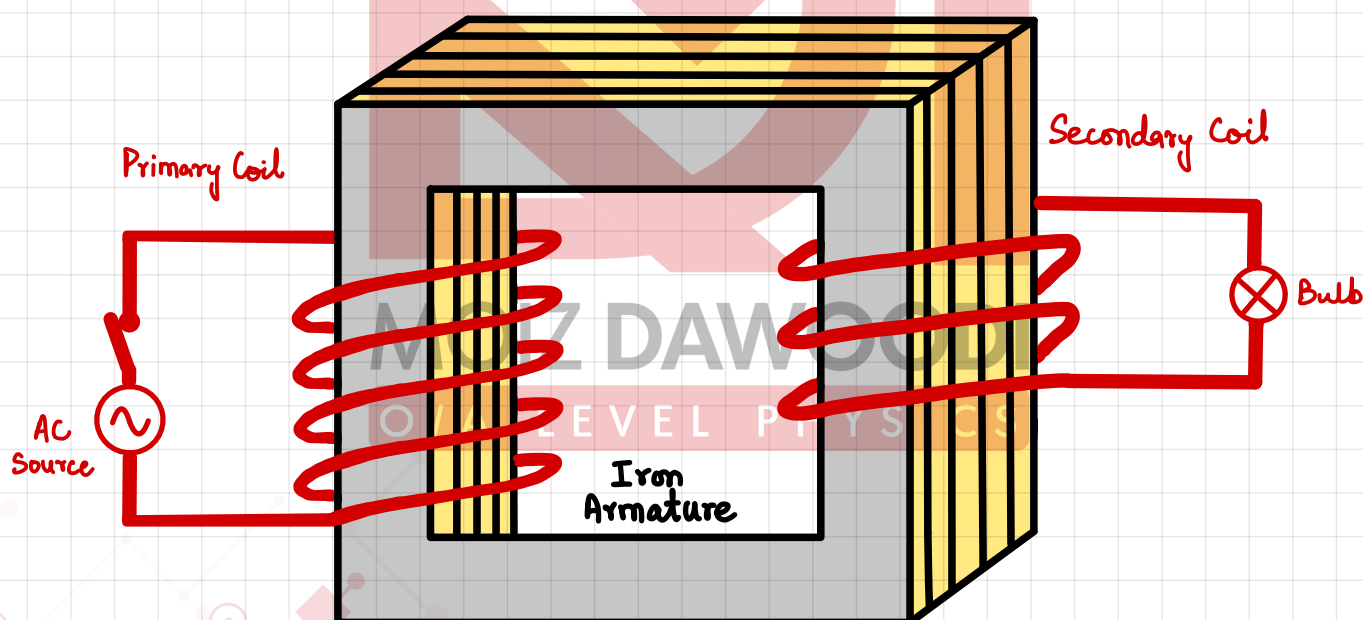
A strong permanent magnet.

Fleming's Left Hand Rule is used when current is known & force is unknown.

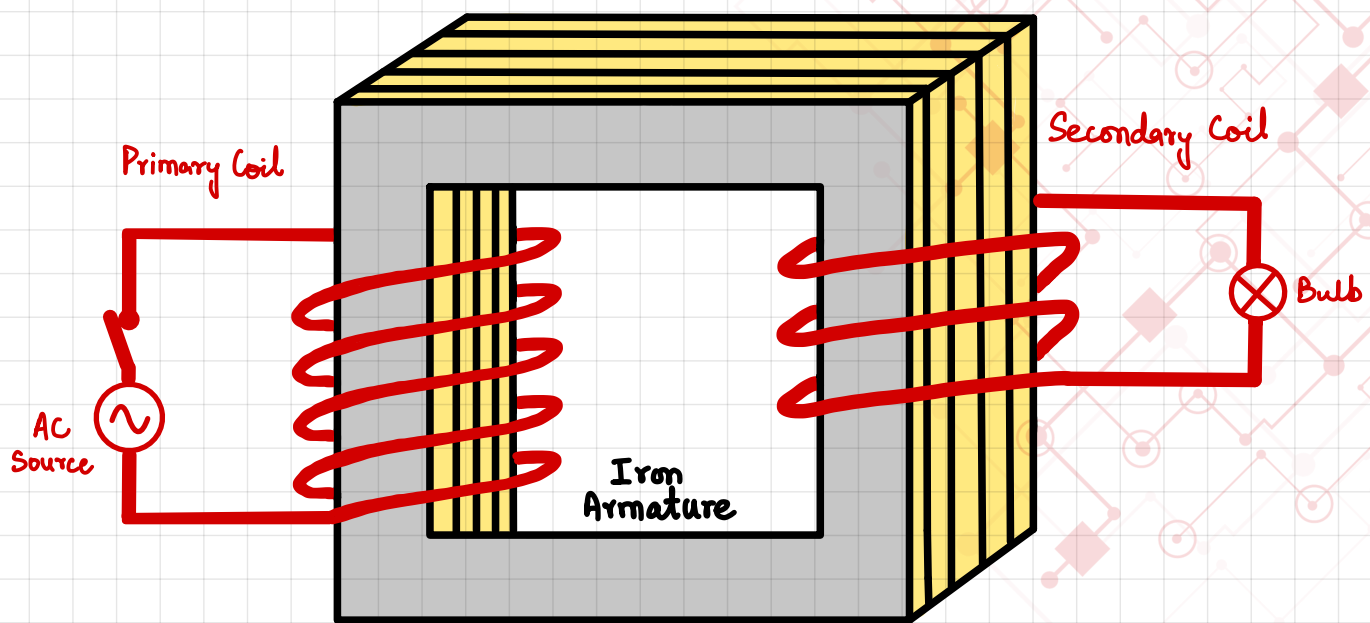
4.5.6

The Transformer

TRANSFORMER:



When positive cycle of AC goes to primary coil, it magnetises & generates its north pole upward & south pole downward. Magnetic field reaches secondary coil & its north pole appears downward while its south pole appears upward.

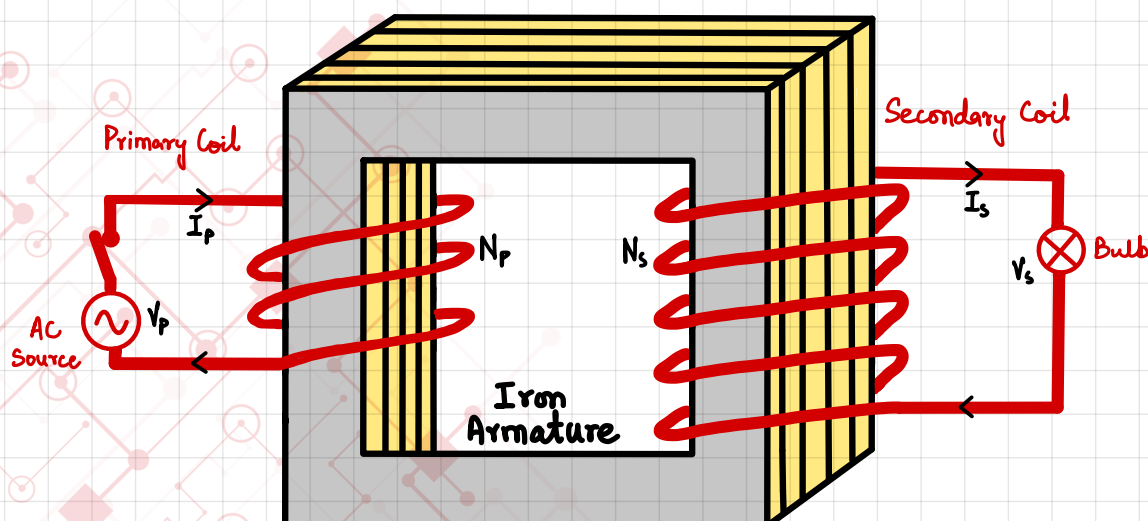


When negative cycle of AC goes to primary coil, it magnetises & generates its north pole downward & south pole upwards. Magnetic field reaches secondary coil & its north pole appears upward while its south pole appears downward. So when the current reverses at primary coil, direction of current also reverses at secondary coil.

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Transformer can be classified as:

STEP-UP TRANSFORMER:

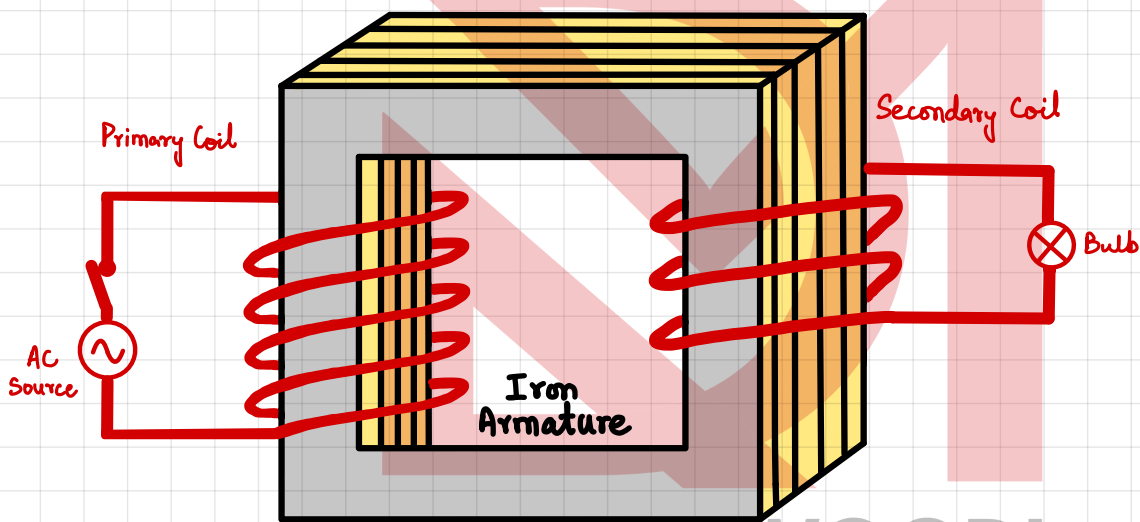


N_p = No. of turns of Primary Coil
 I_p = Current of Primary Coil
 V_p = Voltage of primary coil
 N_s = No. of turns of Secondary Coil
 I_s = Current of Secondary Coil
 V_s = Voltage of Secondary Coil

Draw a diagram of Step Up Transformer.

N_p	N_s , Turns
V_p	V_s , Voltage
L_p	L_s , Length
R_p	R_s , Resistance
I_p	I_s , Current

STEP-DOWN TRANSFORMER:



N_p = No. of turns of Primary Coil
 I_p = Current of Primary Coil
 V_p = Voltage of primary coil
 N_s = No. of turns of Secondary Coil
 I_s = Current of Secondary Coil
 V_s = Voltage of Secondary Coil

Draw a diagram of Step Up Transformer.

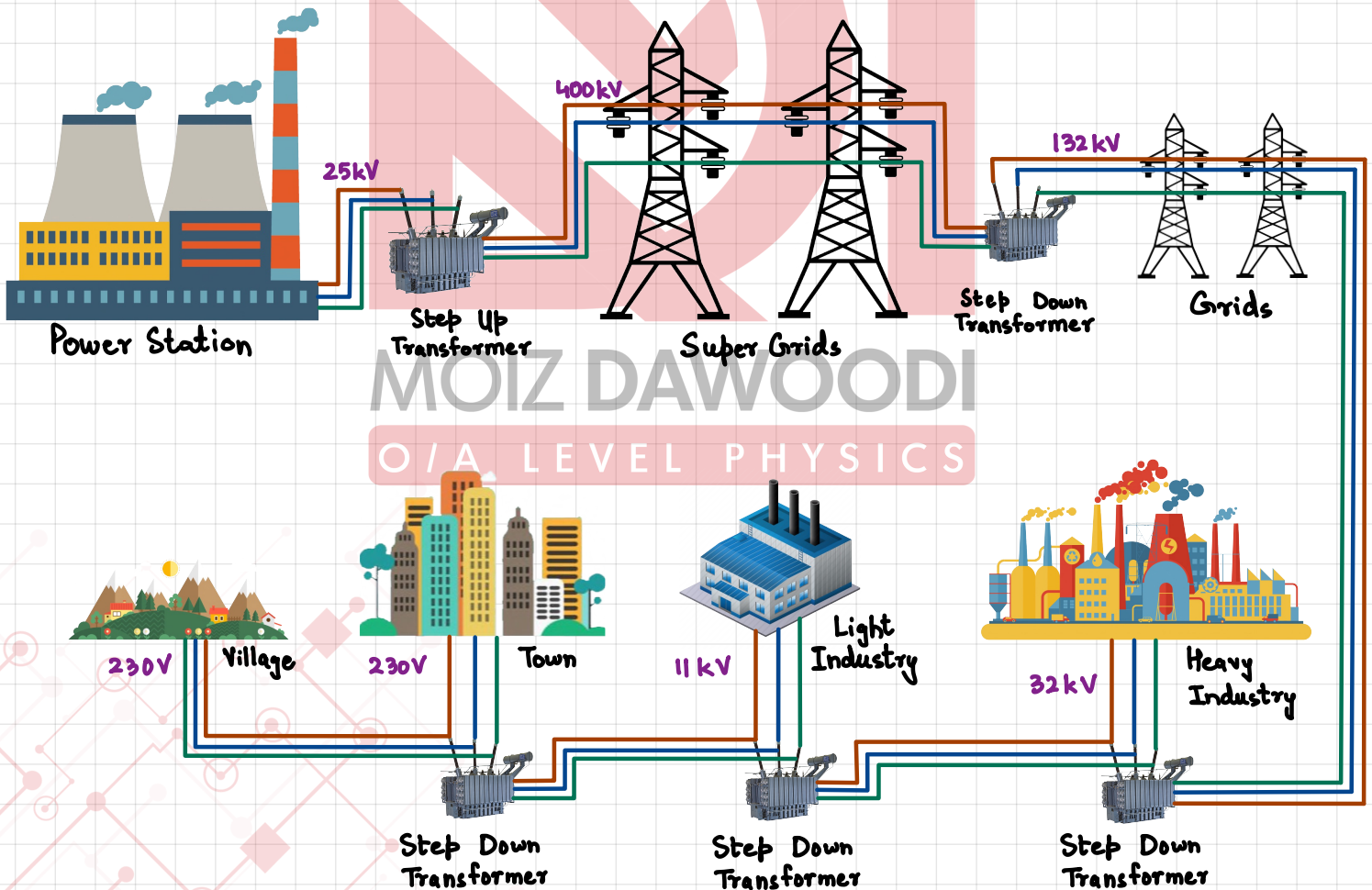
N_p	N_s , Turns
V_p	V_s , Voltage
L_p	L_s , Length
R_p	R_s , Resistance
I_p	I_s , Current

FORMULATION:

If a transformer is 100% efficient,
So:

So, we can write as:

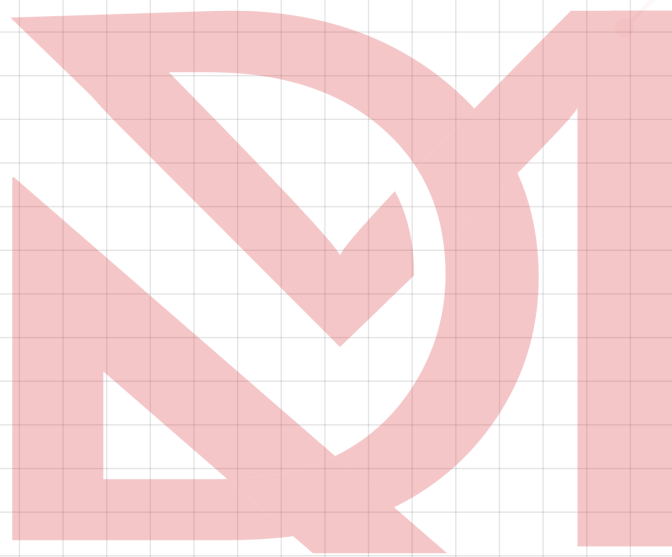
TRANSMISSION THROUGH POWER LINES:



HOW TO REDUCE POWER LOSS:

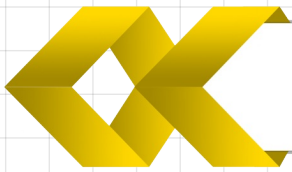
Q: Why electricity is transmitted at high voltages?

A

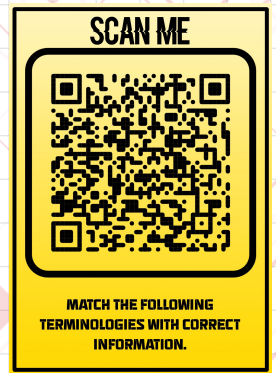


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Match the following



Like current wires

converts low voltage AC to high voltage AC.

Faraday's law says

directly proportional to emf induced.

Unlike current wires

attract each other.

Step-up Transformer

change of flux causes induced emf.

No. of turns of coil is

Lets current to reverse its direction.

Solenoid has same

repel each other

Split ring commutator

represent outward magnetic field.

Cross 'x' represents

shape of magnetic field as bar magnet.

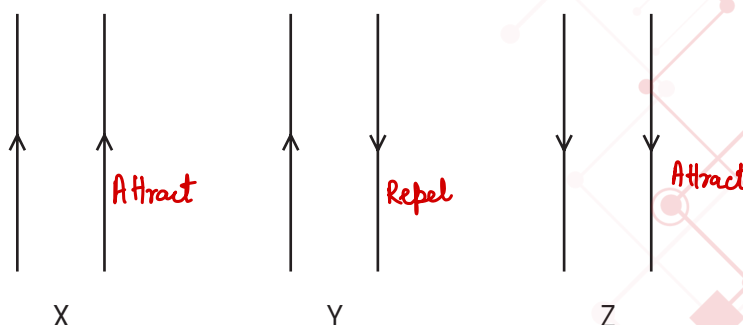
Multiple dots '•'

by moving magnet at faster rate.

Flux can be increased

inward current.

01 The diagram shows three pairs of parallel wires with the currents in the directions shown.



For each pair of wires, what are the forces between the wires?

	X	Y	Z
A	attraction ✓	none	repulsion
B	attraction ✓	repulsion ✓	attraction ✓
C	repulsion	attraction	repulsion
D	repulsion	repulsion ✓	repulsion

[MJ2011/P11/Q33]

02 The coil in an electric motor is wound onto a cylinder.

Why is the cylinder made of soft iron?

- ☒ A to deflect the magnetic field away from the coil
- ☒ B to increase the current through the coil
- ☐ C to increase the strength of the magnetic field through the coil
- ☒ D to support the coil and prevent it from collapsing

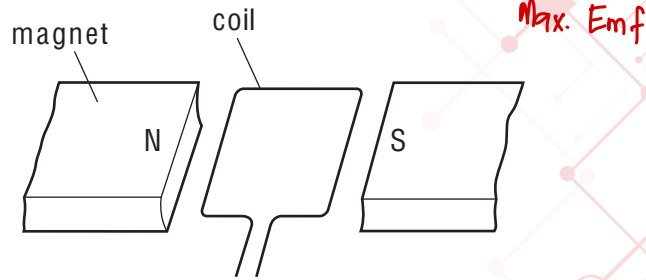
[MJ2011/P11/Q34]

03 The electromotive force (e.m.f.) induced in a conductor moving at right-angles to a magnetic field does **not** depend upon

- ☐ A the length of the conductor.
- ☒ B the resistance of the conductor. ✓
- ☒ C the speed of the conductor. ✓
- ☒ D the strength of the magnetic field. ✓

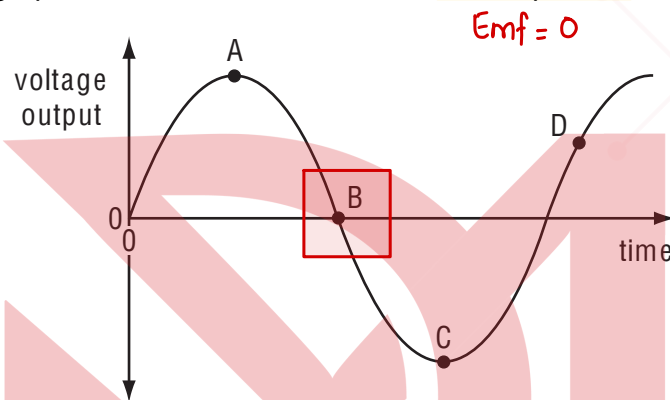
[MJ2011/P11/Q35]

- 04 The diagram shows part of an a.c. generator when its coil is in a horizontal position.



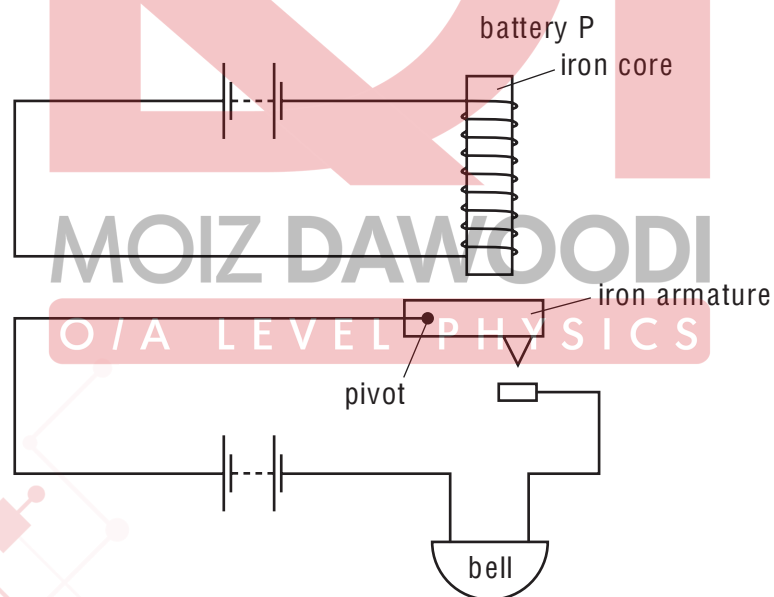
The graph shows the voltage output plotted against time.

Which point on the graph shows when the coil is in a vertical position?



[MJ2011/P11/Q35]

- 05 The diagram shows an alarm system.



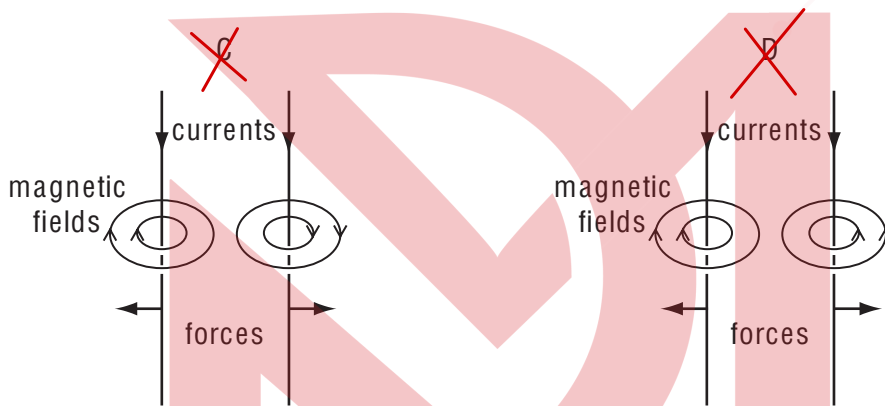
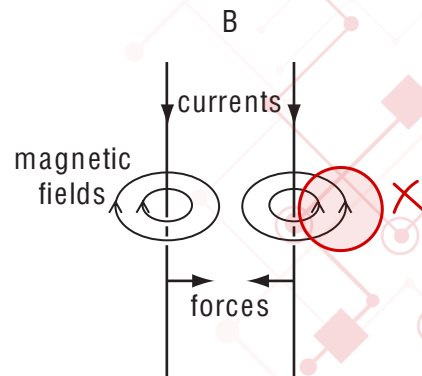
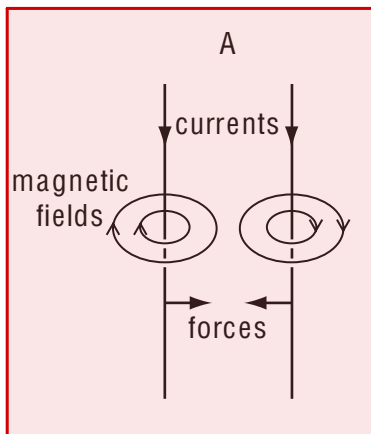
What happens when battery P is disconnected?

	iron armature	bell
A	falls ✓	rings ✓
B	falls ✓	stops ringing
C	moves up	rings ✓
D	moves up	stops ringing

[ON2011/P11/Q25]

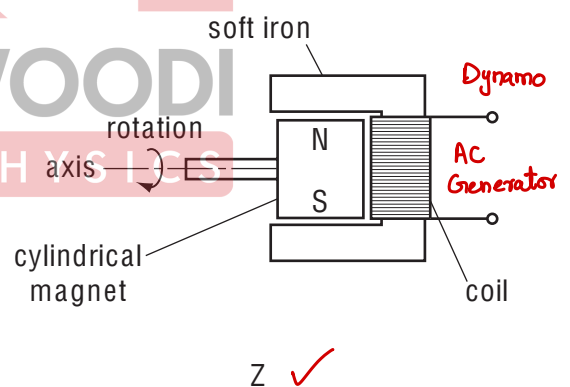
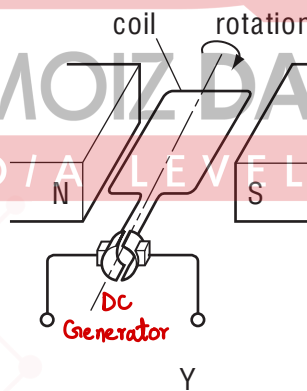
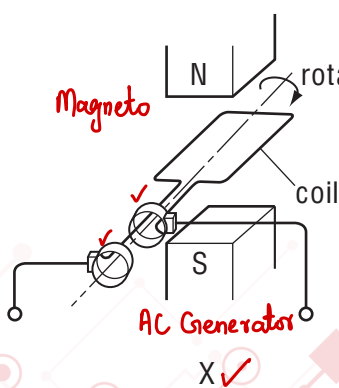
06 Two parallel wires carry currents in the same direction.

Which diagram shows the magnetic field around each wire and the direction of the force on each wire?



[ON2011/P11/Q34]

07 The diagrams show three electrical devices, X, Y and Z.



Which devices provide an alternating current (a.c.) output?

A X only

B Y only

C X and Y

D X and Z

[ON2011/P11/Q35]

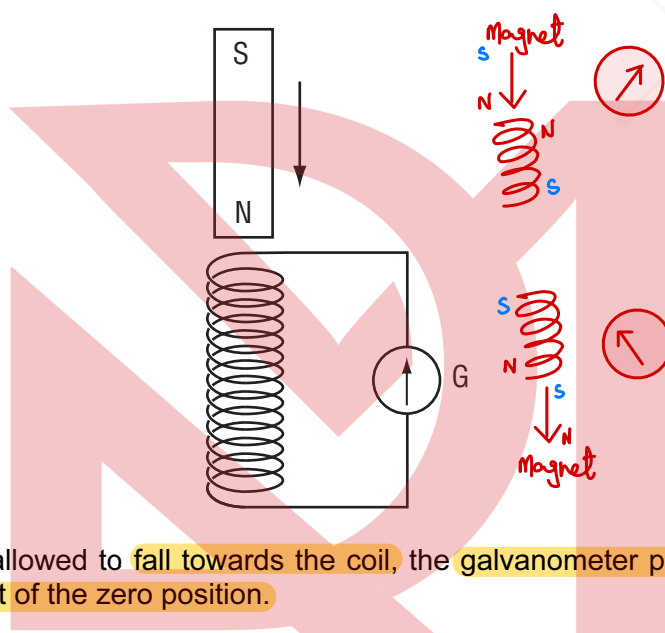
08 One component of a simple d.c. motor is a split-ring commutator.

Which metal is used to make the commutator, and why is this metal chosen?

	metal	reason
A	copper ✓	it is a good conductor of electricity ✓
B	copper ✓	it is a good conductor of heat
C	iron	it increases the magnetic field strength
D	iron	it is attracted to the brushes

[ON2011/P11/Q36]

09 A small coil is connected to a galvanometer G, as shown.



When a magnet is allowed to fall towards the coil, the galvanometer pointer gives a momentary deflection to the right of the zero position.

The magnet moves through the coil.

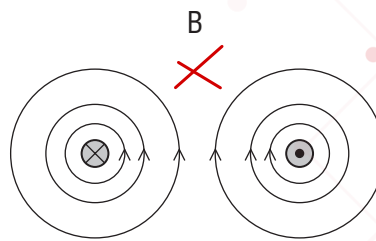
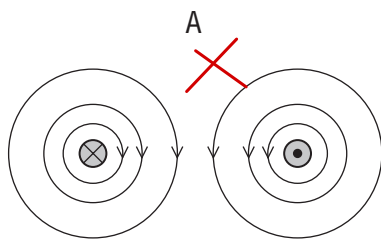
What happens to the galvanometer pointer as the magnet falls away from the coil?

- ☒ A It gives a continuous reading to the left.
- ☐ B It gives a momentary deflection to the left.
- ☒ C It gives a continuous reading to the right.
- ☒ D It gives a momentary deflection to the right.

[ON2011/P11/Q37]

- 10 Two straight electrical conductors are parallel to one another. Each carries a current, one into the plane of the paper and one out of the plane of the paper.

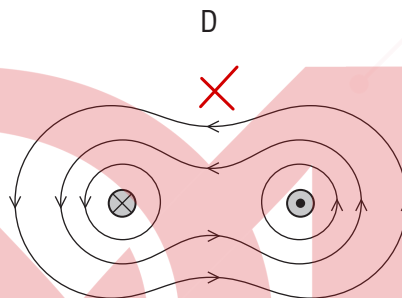
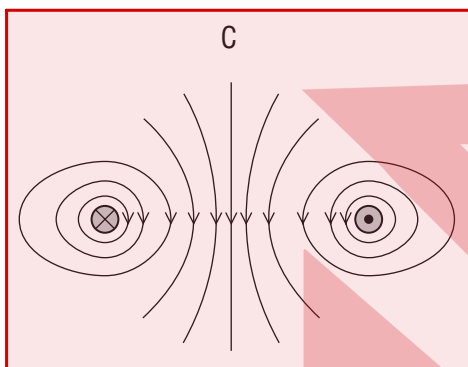
Which diagram shows the magnetic field around the two wires?



key

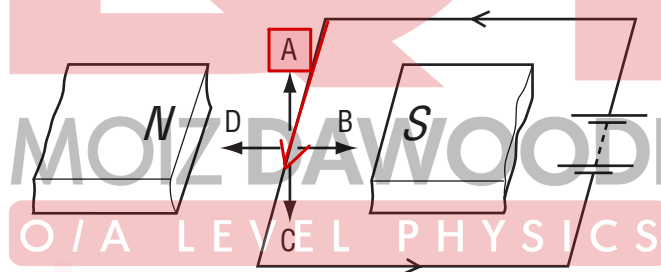
⊗ current into plane of paper

⊙ current out of plane of paper



[MJ2012/P11/Q26]

- 11 A current-carrying wire is placed between the poles of a magnet. What is the direction of the force on the wire due to the current?



[MJ2012/P11/Q33]

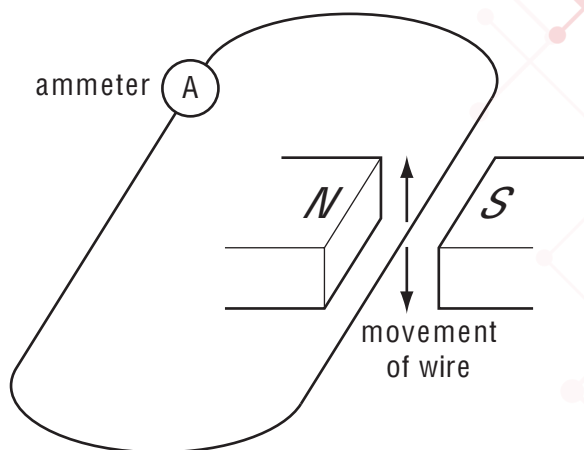
- 12 A simple d.c. motor consists of a coil that rotates between the poles of a permanent magnet. The turning effect is increased by winding the coil on a metal cylinder.

Which metals are used to make the magnet and the cylinder?

	magnet	cylinder
A	iron	copper
B	iron	steel
C	steel ✓	copper
D	steel ✓	iron ✓

[MJ2012/P11/Q34]

- 13 A current is produced when a wire is moved between two magnets as shown.



Which device uses this effect?

- A a battery
- B a generator**
- C a motor
- D an electromagnet

[MJ2012/P11/Q35]

- 14 Why is a transformer used to connect a generator in a power station to a long-distance transmission line?

- A to decrease the voltage and decrease the current
- B to decrease the voltage and increase the current
- C to increase the voltage and decrease the current
- D to increase the voltage and increase the current

[MJ2012/P11/Q36]

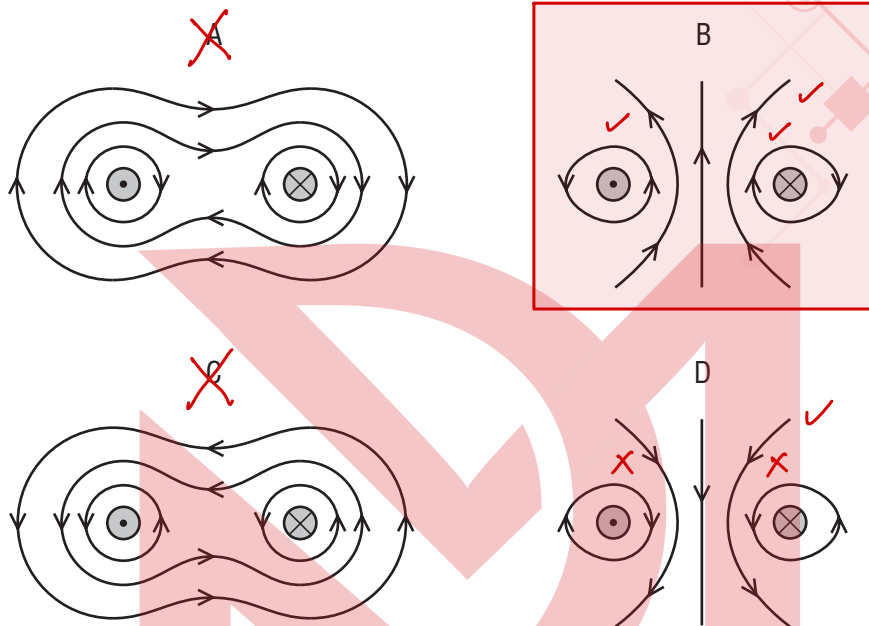
MOIZ DAWOODI
O / A LEVEL PHYSICS

- 15 Two long, straight wires hang vertically, close to each other. The wires carry currents in opposite directions.


current
out of page


current
into page

Which diagram shows the magnetic field pattern around the wires?



[MJ2012/P12/Q26]

- 16 The diagram shows a beam of electrons entering a magnetic field. The direction of the field is into the page.

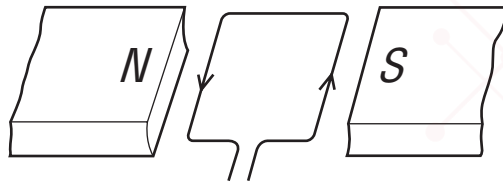


In which direction are the electrons deflected?

- A into the page
- B out of the page
- C towards the bottom of the page**
- D towards the top of the page

[MJ2012/P12/Q33]

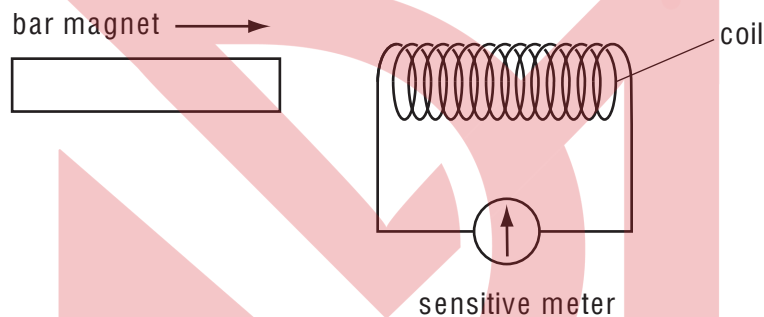
- 17 A rectangular coil is placed between the poles of a magnet. A current passes through the coil, as shown.



What happens to the coil?

- A It moves downwards.
- B It moves upwards.
- C It rotates anticlockwise.
- D It rotates clockwise.

- 18 A bar magnet is pushed into one end of a long coil connected to a sensitive meter.



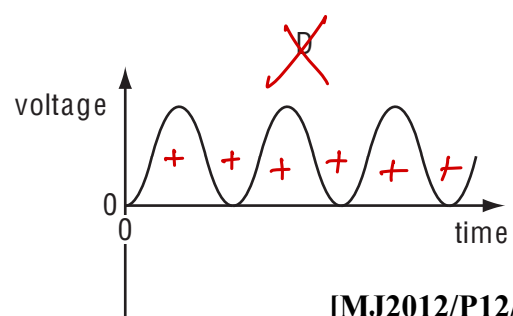
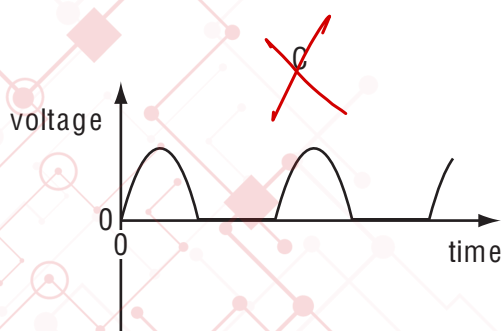
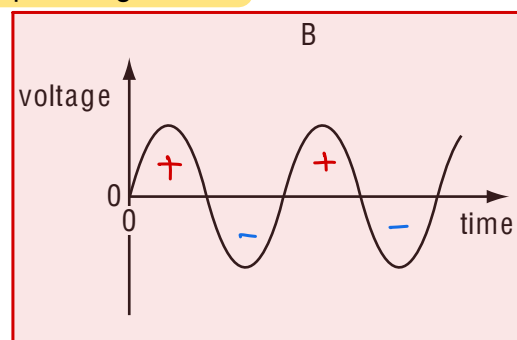
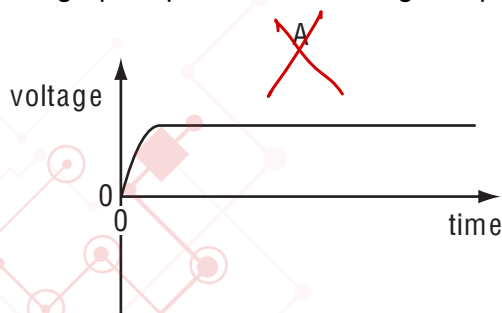
[MJ2012/P12/Q34]

Which of the following affects the **magnitude** of the **deflection of the meter**?

- A the direction in which the coil is wound
- B the speed with which the magnet enters the coil
- C which end of the coil is used
- D which pole of the magnet enters first

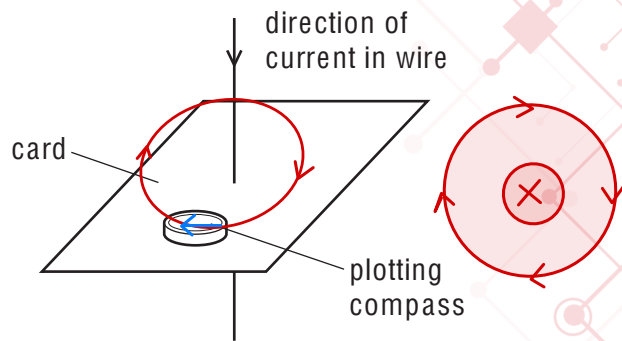
[MJ2012/P12/Q35]

- 19 Which graph represents the voltage output of a simple a.c. generator?



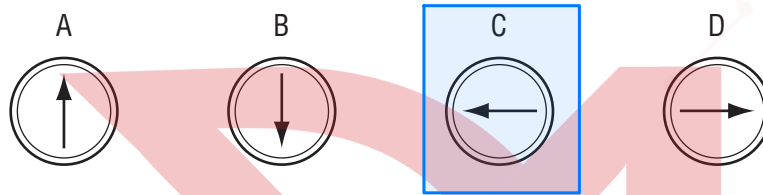
[MJ2012/P12/Q36]

- 20 A vertical wire passes at right angles through a piece of card. There is a large current in the wire in the direction shown.



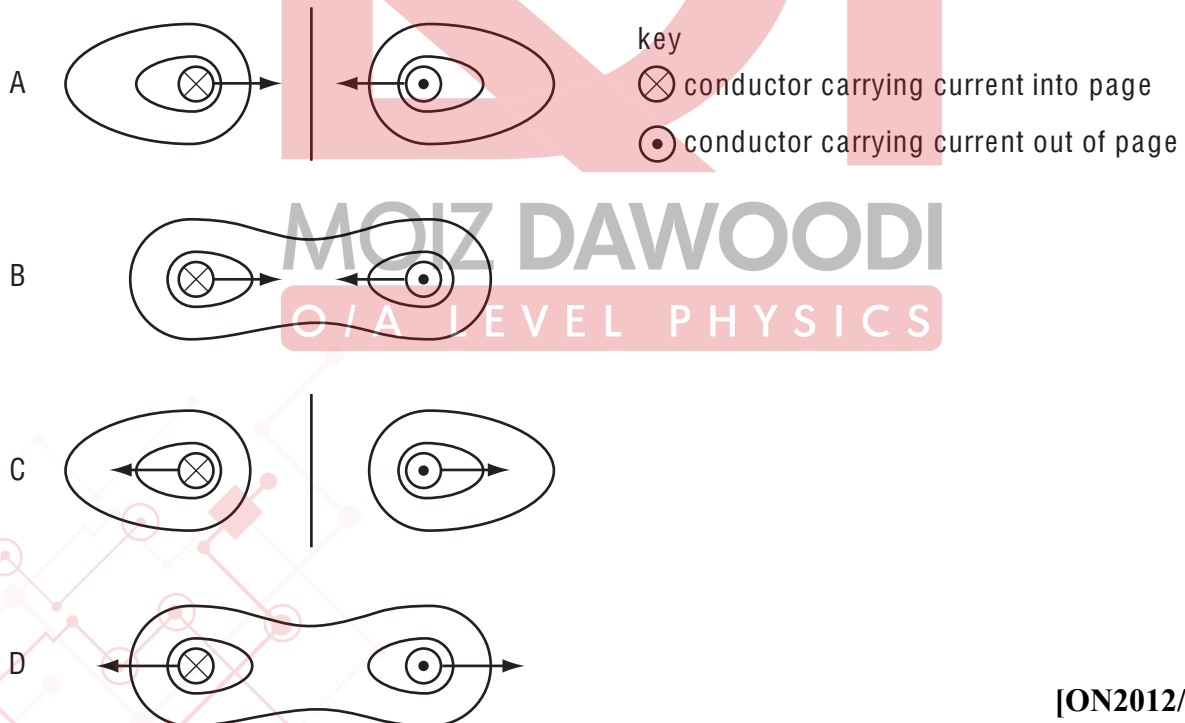
A plotting compass is placed on the card.

Which diagram shows the direction in which the needle of the plotting compass points?



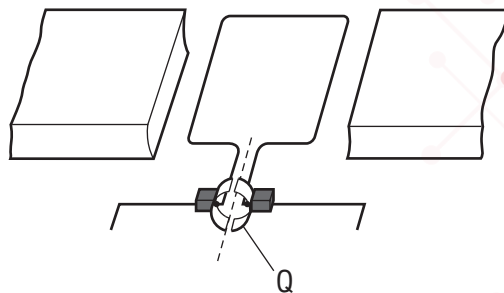
[ON2012/P11/Q26]

- 21 Each of the diagrams shows a cross-section through two parallel, current-carrying conductors. Which diagram shows the shape of the magnetic field pattern **and** the directions of the forces on the two conductors?



[ON2012/P11/Q33]

22 The diagram shows a simple d.c. motor.

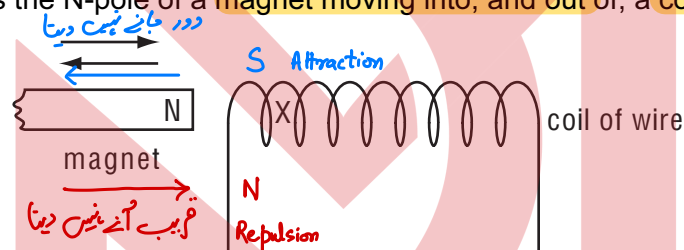


What is the part labelled Q?

- A a coil
- B a magnet
- C a slip ring
- D a split-ring commutator

[ON2012/P11/Q34]

23 The diagram shows the N-pole of a magnet moving into, and out of, a coil of wire.



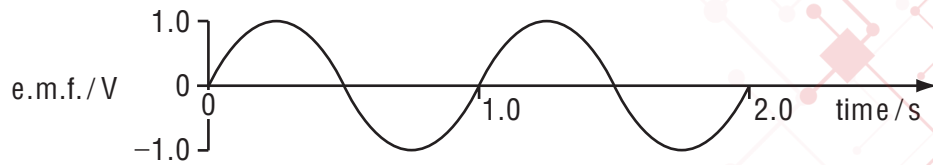
This movement produces a current in the coil of wire. The current produces a magnetic pole at X.

Which pole is produced at X when the magnet is moved in and when it is moved out?

	magnet moved in	magnet moved out
A	N ✓	N
B	N ✓	S ✓
C	S	N
D	S	S ✓

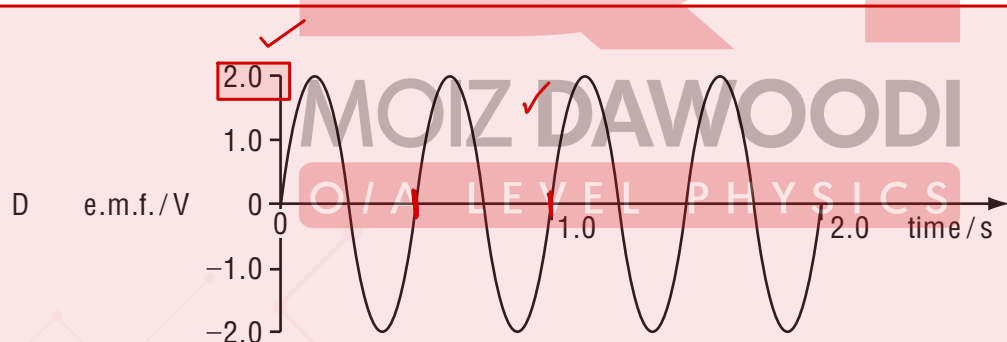
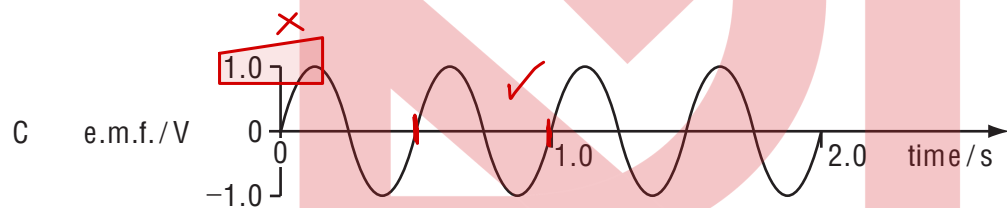
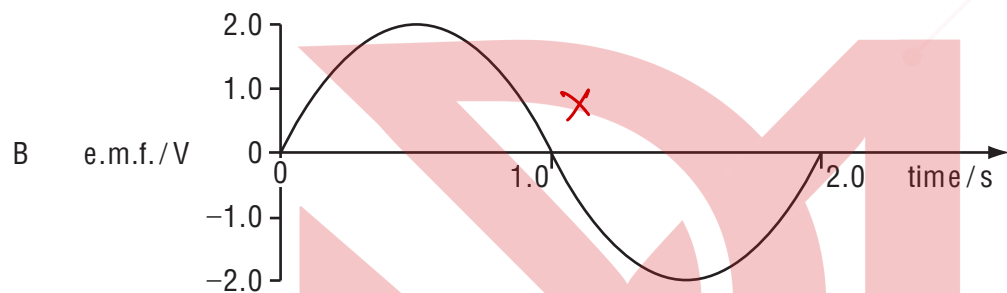
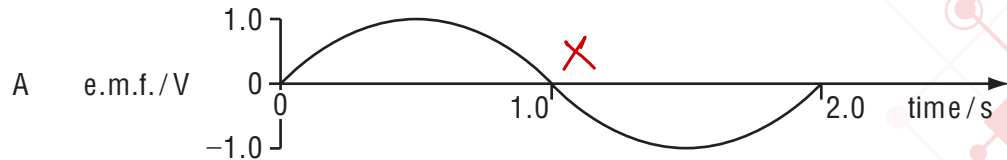
[ON2012/P11/Q35]

24 A simple a.c. generator produces an alternating e.m.f. as shown.



The speed of the generator is doubled. $2 \text{ emf} \propto 2 \text{ V}$

Which graph best represents the new output? $2f \propto 2 \text{ V}$



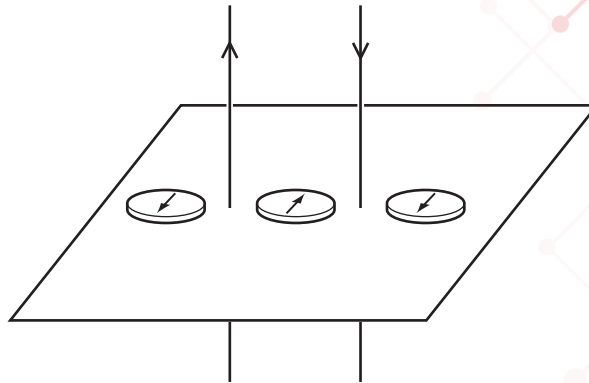
[ON2012/P11/Q36]

25 What is required to operate a reed relay in a switching circuit?

- ☒ A a capacitor
- ☐ B an electric field
- ☒ C a magnetic field
- ☐ D a transformer

[ON2012/P11/Q37]

- 26 Two parallel wires carry currents in opposite directions. Three plotting compasses are placed in the positions shown.



[ON2012/P12/Q33]

The currents in **both** wires are reversed. How many compass needles change direction? (Ignore the effect of the Earth's magnetic field.)

- A 0 B 1 C 2 D 3

- 27 Which single-coil motor has the largest turning effect?

	current in coil / A	number of turns in coil	iron core
A	6	100	no
B	10	200	no
C	6	100	yes
D	10	200	yes

[ON2012/P12/Q34]

- 28 A magnet is moved towards a coil of insulated wire. A voltmeter connected across the coil shows a positive reading.

What produces a higher reading on the voltmeter?

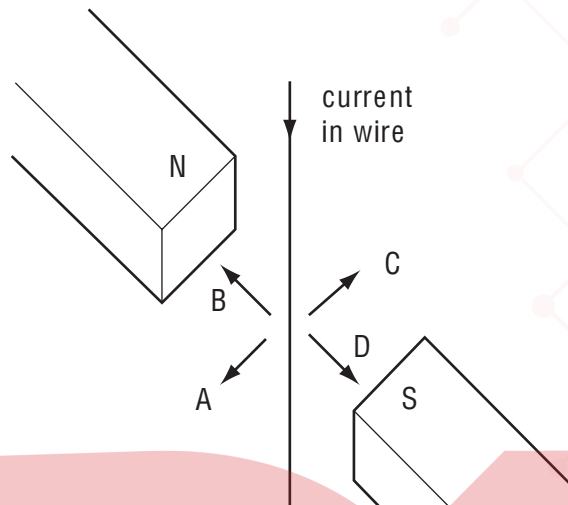
- ☒ A moving the magnet away from the coil at the same speed
☒ B moving the magnet away from the coil at a slower speed
☒ C moving the magnet towards the coil at a faster speed
☒ D moving the magnet towards the coil at a slower speed

N → + Reading

[ON2012/P12/Q35]

29 A wire hangs between the poles of a magnet.

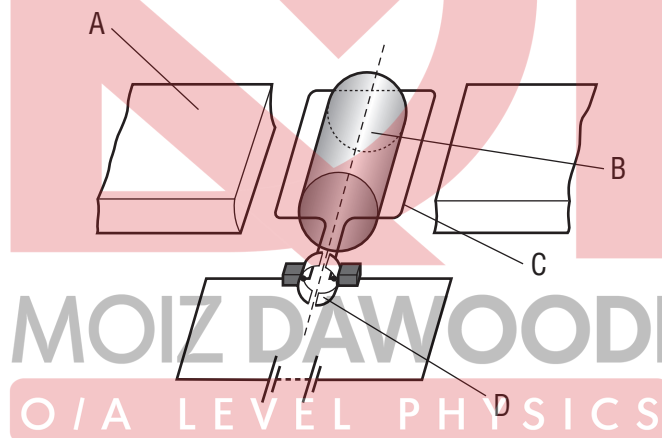
When there is a current in the wire, in which direction does the wire move?



[MJ2013/P11/Q34]

30 The diagram shows a simple d.c. motor.

Which labelled part is the commutator?



[MJ2013/P11/Q35]

31 Two leads emerging from a box are connected to a sensitive ammeter.



When a bar magnet moves towards the open end of the box, the needle of the ammeter deflects to the right. When the bar magnet stops, the needle returns to zero.

What is inside the box?

A a coil alone

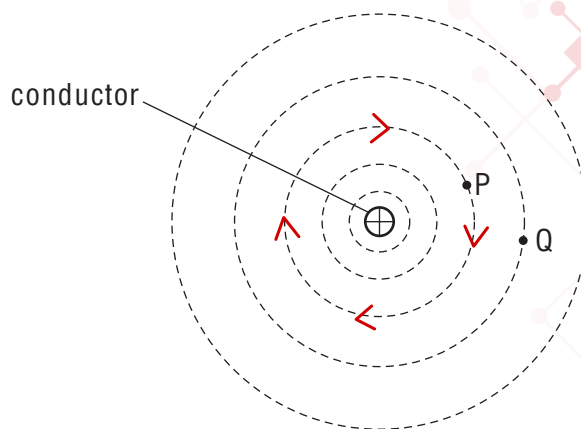
B a coil connected in series with a cell

~~C~~ a light-dependent resistor (LDR) alone

~~D~~ an LDR in series with a cell

[MJ2013/P11/Q36]

32 The diagram shows the shape of the magnetic field lines near a current-carrying conductor.



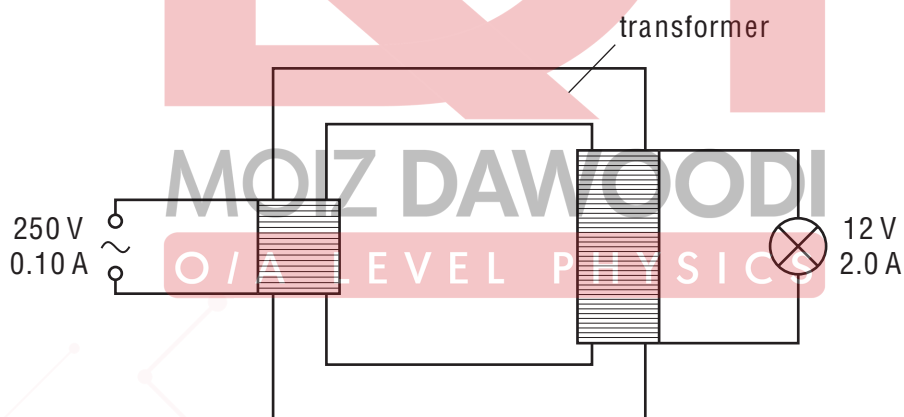
The current in the conductor is into the plane of the diagram.

Which row correctly states the direction of the field lines and compares the strengths of the field at points P and Q?

	direction of field lines	the field is stronger at
A	clockwise ✓	P ✓
B	clockwise ✓	Q
C	anticlockwise	P ✓
D	anticlockwise	Q

[ON2013/P11/Q25]

33 A transformer is used to operate a 12V lamp from a 250V mains supply.



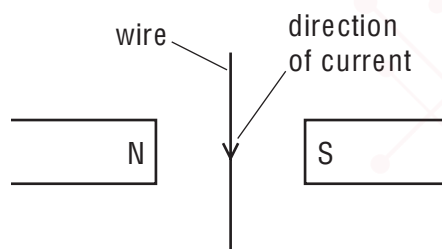
The mains current is 0.10 A. The current in the lamp is 2.0 A.

What is the efficiency of the transformer?

- A** 0.048 **B** 0.050 **C** 0.96 **D** 1.04

[ON2013/P11/Q31]

34 A current-carrying wire lies between the poles of two magnets, as shown.



What is the direction of the force on the wire?

- A into the plane of the paper
- B out of the plane of the paper
- C towards the N-pole
- D towards the S-pole

[ON2013/P11/Q32]

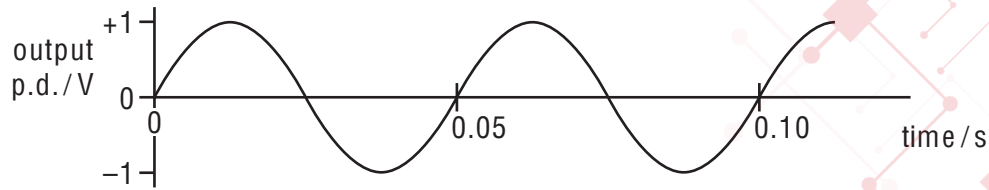
35 Which material is used for the core of a transformer and why?

	material	reason
A	copper	good conductor of electricity
B	copper	easy to magnetise and demagnetise
C	iron	good conductor of electricity
D	iron	easy to magnetise and demagnetise

[ON2013/P11/Q34]

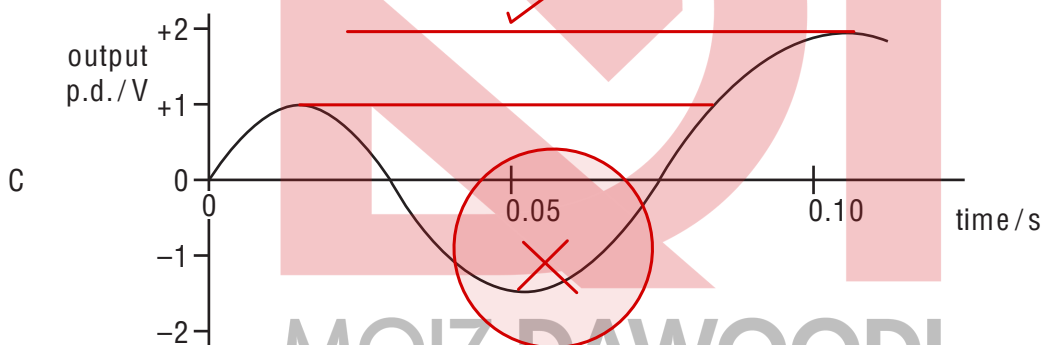
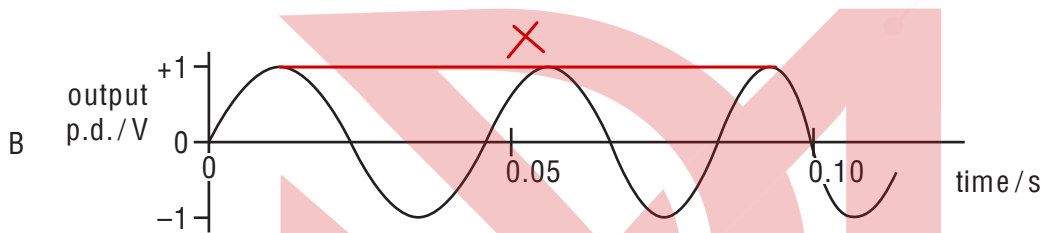
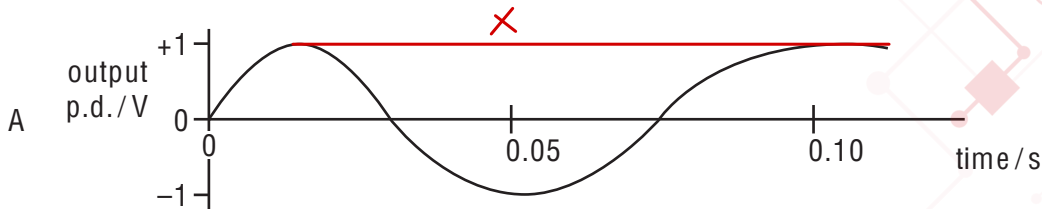
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O / A LEVEL PHYSICS

- 36 The graph shows the output of an a.c. generator. The coil in the generator rotates 20 times in one second.



The speed of rotation of the coil steadily increases.

Which graph best shows how the output changes?



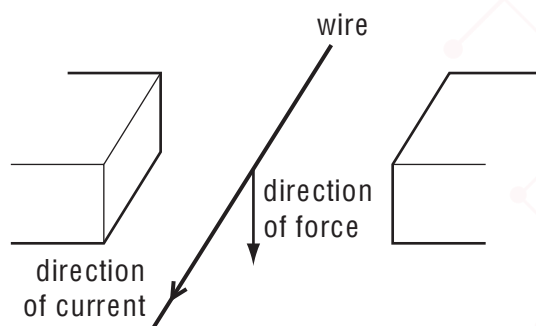
$$\begin{aligned} \uparrow \text{emf} &\propto v \uparrow \uparrow \\ \uparrow f &\propto v \uparrow \uparrow \end{aligned}$$

$$\begin{aligned} \uparrow f &\propto v \uparrow \\ \uparrow f &\propto \frac{1}{\lambda} \downarrow \end{aligned}$$

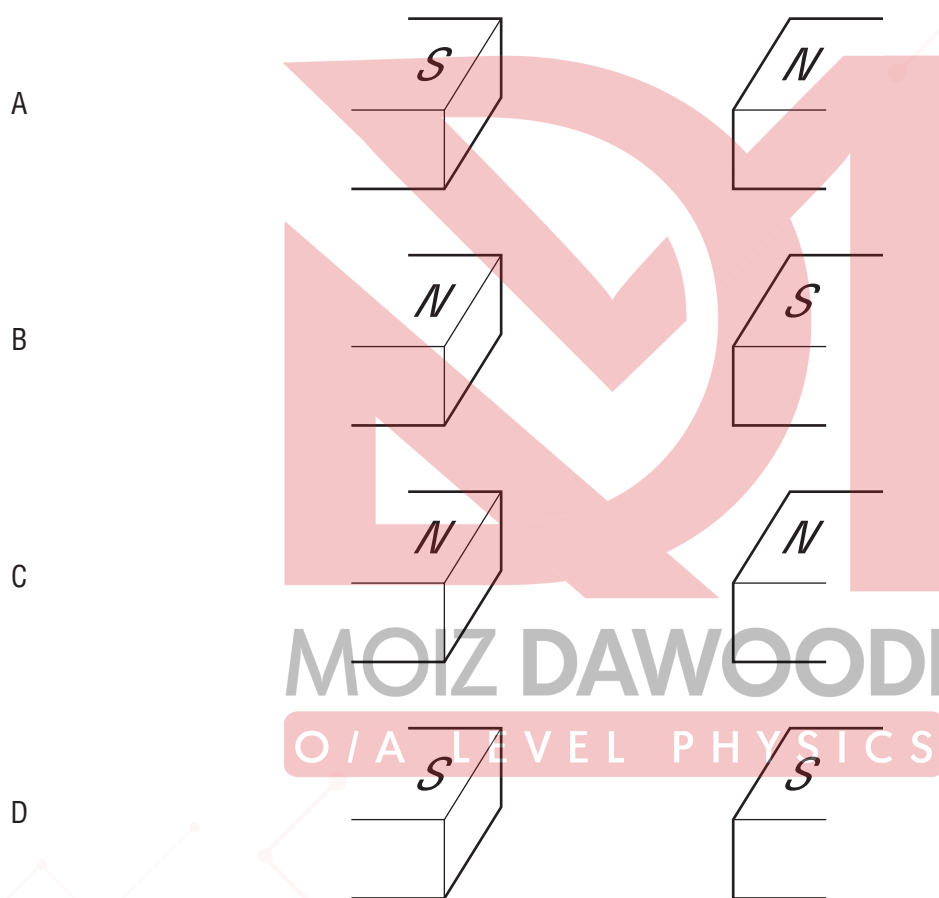
[ON2013/P11/Q33]

37 The diagram shows a wire placed between two magnetic poles of equal strength.

A current passes through the wire in the direction shown. The current causes a downward force on the wire.

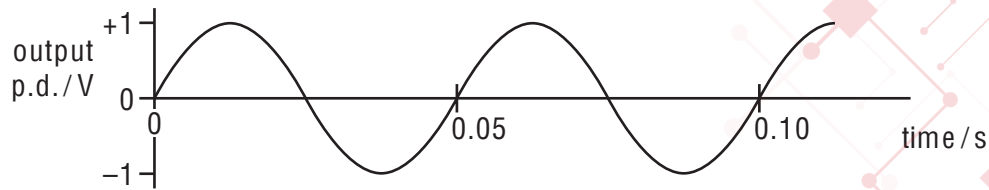


What is the arrangement of the magnetic poles?



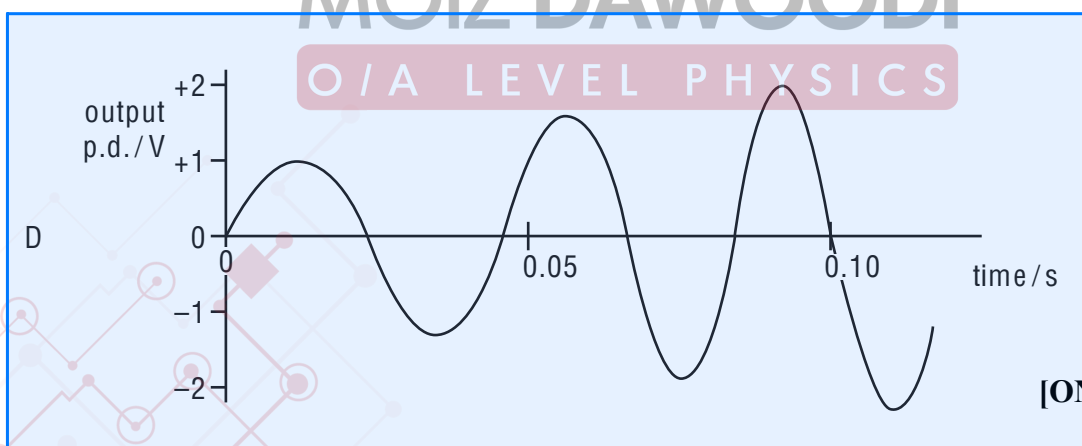
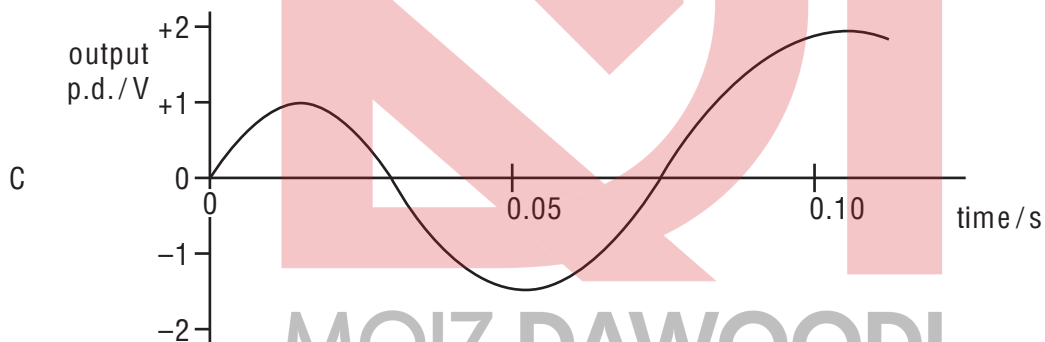
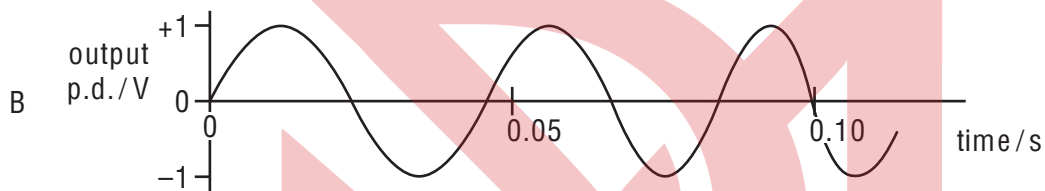
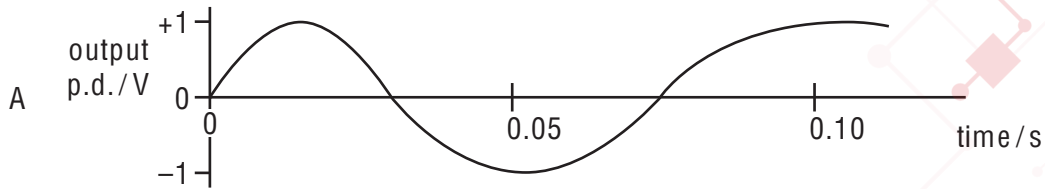
[ON2013/P12/Q32]

- 38 The graph shows the output of an a.c. generator. The coil in the generator rotates 20 times in one second.



The speed of rotation of the coil steadily increases.

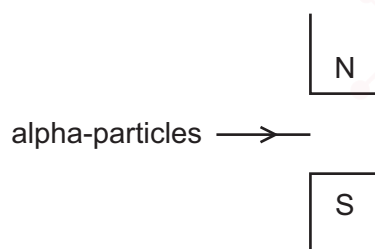
Which graph best shows how the output changes?



Repeated!

[ON2013/P12/Q33]

39 A beam of alpha-particles enters the magnetic field between the poles of a magnet.

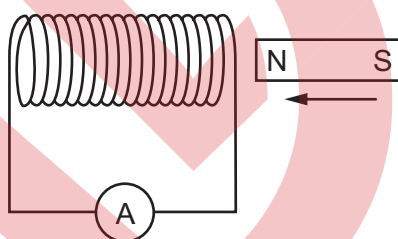


In which direction is the magnetic force on the beam?

- A down the page
- B into the page
- C out of the page
- D up the page

[MJ2014/P11/Q34]

40 As a magnet is moved into the coil of wire as shown, there is a small reading on the sensitive ammeter.

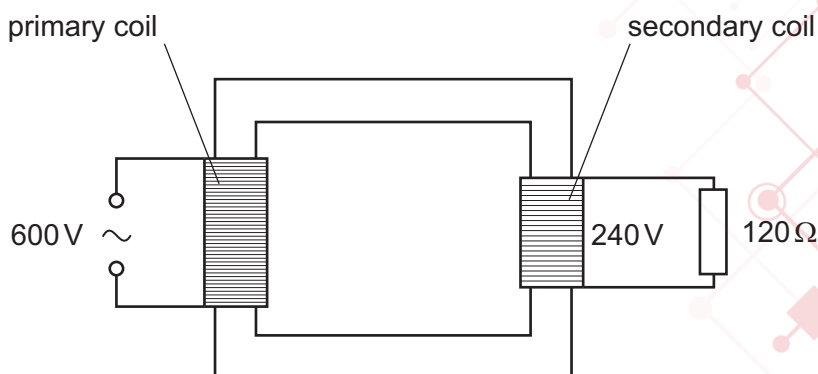


Which change increases the size of the reading?

- ~~A~~ moving the opposite pole into the coil
- ~~B~~ pulling the magnet out of the coil
- C** pushing the magnet in faster
- ~~D~~ unwinding some of the turns of wire

[MJ2014/P11/Q35]

- 41 An ideal transformer has a primary voltage of 600 V and a secondary voltage of 240 V. The secondary coil is attached to a resistor of resistance $120\ \Omega$.



What is the power dissipated in the resistor and the current in the primary coil?

	power / W	current / A
A	120	0.20
B	120	5.0
C	480	0.80
D	480	1.3

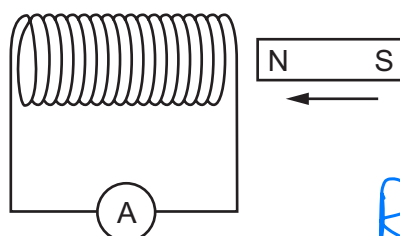
[MJ2014/P11/Q36]

- 42 Which device uses the force experienced by a current in a magnetic field when in normal use?

- A cathode-ray oscilloscope
- B electrostatic precipitator
- C loudspeaker**
- D transformer

[MJ2014/P12/Q34]

- 43 As a magnet is moved into the coil of wire as shown, there is a small reading on the sensitive ammeter.



Repeated

Which change increases the size of the reading?

- A moving the opposite pole into the coil
- B pulling the magnet out of the coil
- C pushing the magnet in faster**
- D unwinding some of the turns of wire

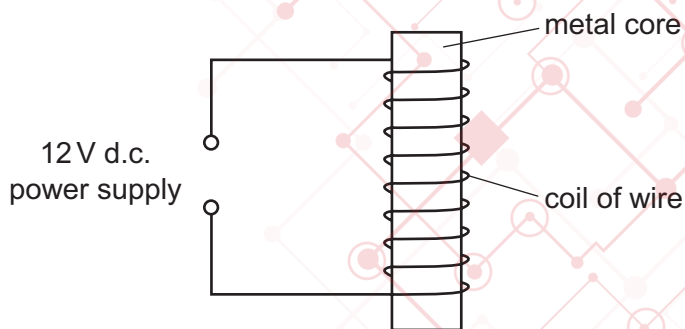
[MJ2014/P12/Q36]

- 44 The diagram shows a **12 V d.c. power supply** connected across a coil with a metal core.

The core becomes a magnet when the current is switched on. It remains a magnet after the current is switched off.

From which metal is the core made?

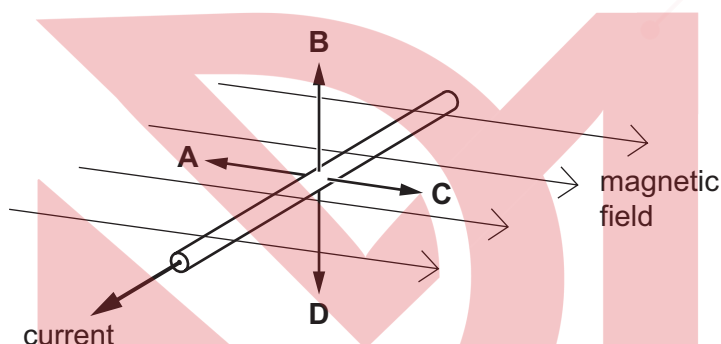
- A aluminium
- B copper
- C soft iron**
- D steel



[ON2014/P11/Q27]

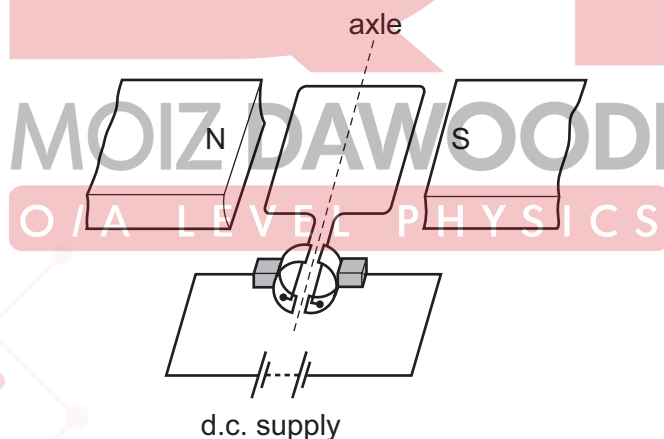
- 45 The diagram shows a current-carrying wire in a horizontal magnetic field.

Which arrow shows the direction of the force experienced by the wire?



[ON2014/P11/Q30]

- 46 The diagram shows a d.c. motor with its coil horizontal.

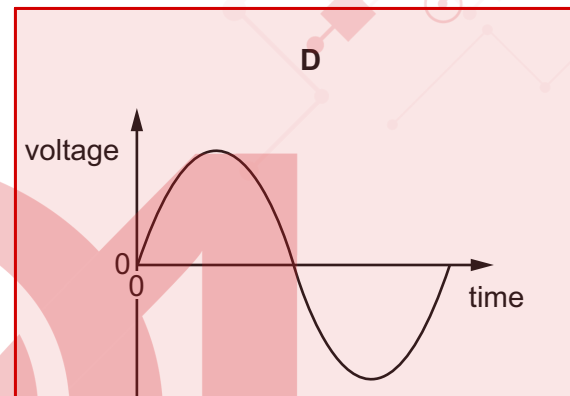
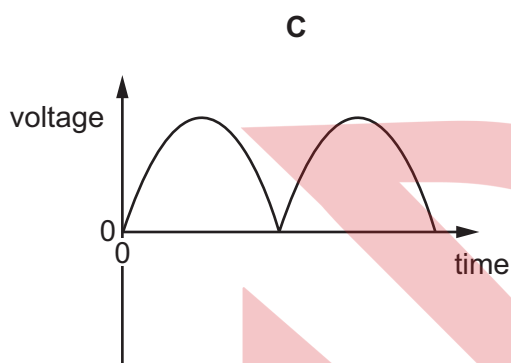
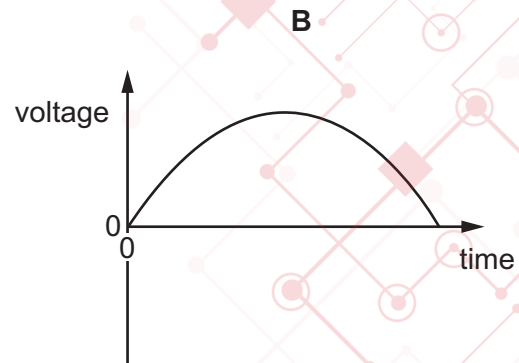
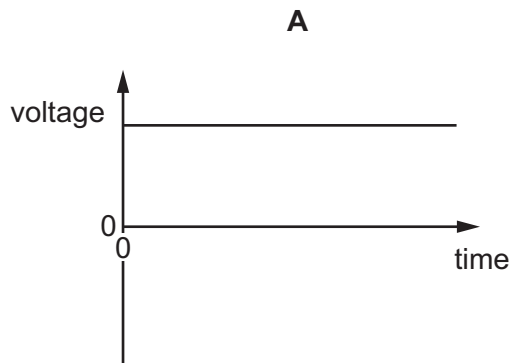


Why is a split-ring commutator used?

- A to change the current direction in the coil as the coil passes the horizontal position
- B to change the current direction in the coil as the coil passes the vertical position
- C to change the current direction in the d.c. supply as the coil passes the horizontal position
- D to change the current direction in the d.c. supply as the coil passes the vertical position

[ON2014/P11/Q31]

47 Which graph shows the voltage output of an a.c. generator when the coil makes **one complete revolution**?



[ON2014/P11/Q32]

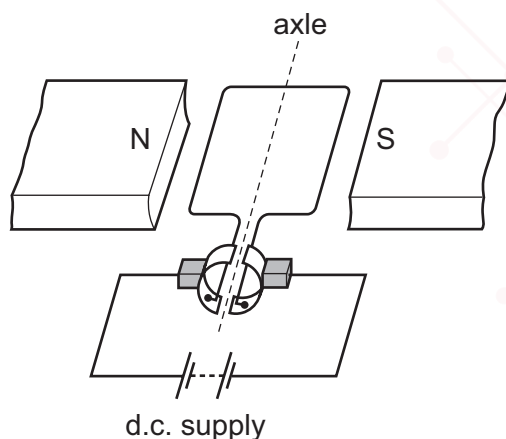
48 Two long, parallel conductors carrying current lie in a horizontal plane. The two conductors attract one another.

The two currents **must**

- A** be in the same direction.
- B** be in opposite directions.
- C** be parallel to the Earth's magnetic field.
- D** be at 90° to the Earth's magnetic field.

[ON2014/P12/Q31]

49 The diagram shows a d.c. motor with its coil horizontal.

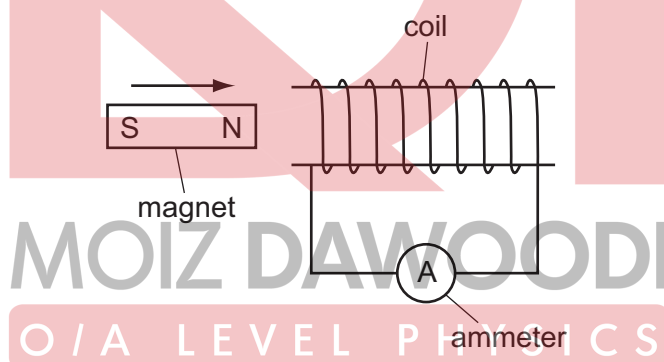


Why is a split-ring commutator used?

- A to change the current direction in the coil as the coil passes the horizontal position
- B to change the current direction in the coil as the coil passes the vertical position
- C to change the current direction in the d.c. supply as the coil passes the horizontal position
- D to change the current direction in the d.c. supply as the coil passes the vertical position

[ON2014/P12/Q32]

50 A student moves a magnet into a coil of wire as shown in the diagram. The coil of wire is connected to a sensitive ammeter.



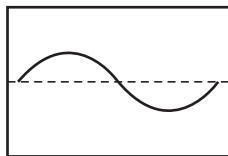
Which change **does not** produce an increase in the reading?

- A increasing the number of turns on the coil ✗
- B increasing the resistance of the ammeter ✓
- C increasing the speed of the magnet ✗
- D increasing the strength of the magnet ✗

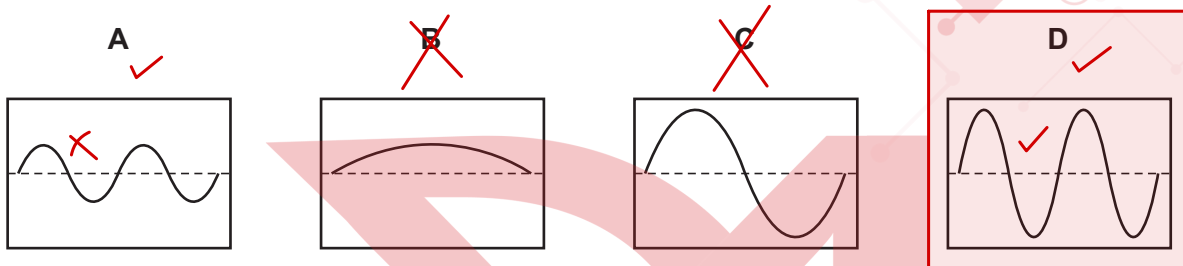
[ON2014/P12/Q33]

- 51 The coil of an **a.c. generator** is rotated and the output is displayed on the screen of a cathode-ray oscilloscope (c.r.o.).

The diagram shows the trace on the screen.

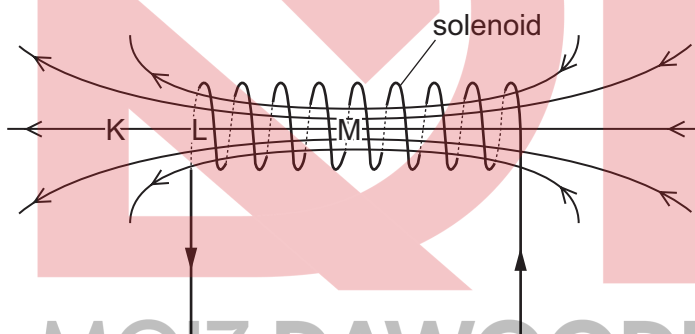


Which trace appears on the screen when the **speed of rotation** of the coil is **doubled** but the settings on the c.r.o. are unaltered?



[ON2014/P12/Q34]

- 52 The diagram shows the magnetic field pattern of a current in a solenoid.

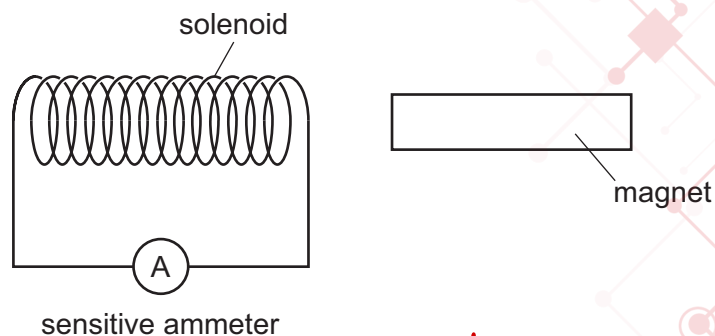


When the **current in the solenoid is increased**, where is there an increase in the magnetic field strength?

- A** K, L and M
- B** K and L only
- C** M and L only
- D** M only

[MJ2015/P11/Q27]

- 53 A magnet is placed near to a solenoid that is connected to a sensitive centre-zero ammeter.



The magnet is pushed towards the solenoid. It accelerates, then moves at constant speed, then decelerates and stops inside the solenoid.

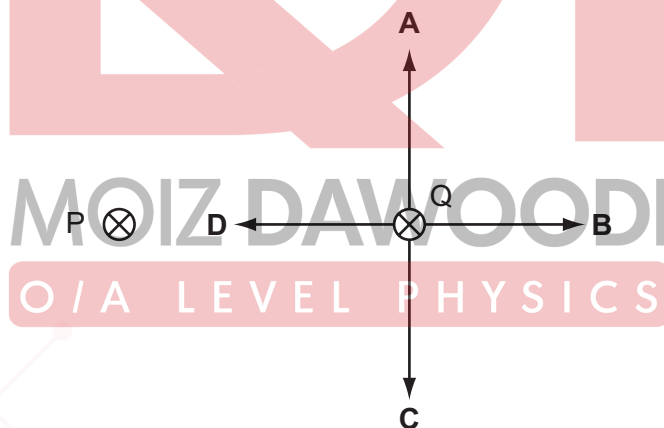
When is the reading on the ammeter zero?

- A when the magnet is accelerating
- B when the magnet is moving at constant speed
- C when the magnet is decelerating
- D when the magnet is stationary

[MJ2015/P11/Q36]

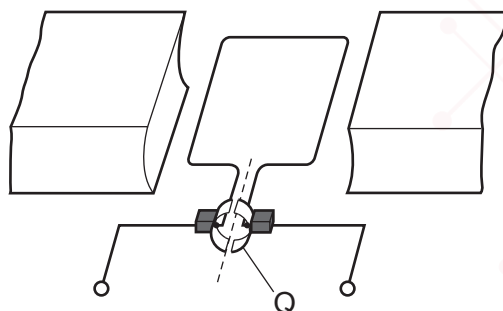
- 32 P and Q represent two, parallel, straight wires carrying currents into the plane of the paper. P and Q exert a force on each other.

Which arrow shows the force on Q?



[ON2015/P11/Q32]

33 The diagram shows a simple d.c. motor.



What is the part labelled Q?

- A a coil
- B a commutator
- C a magnet
- D a slip ring

[ON2015/P11/Q33]

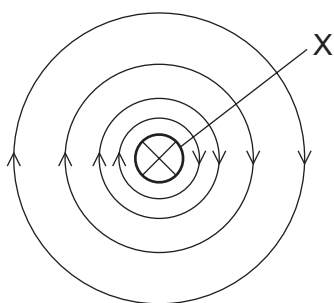
35 Electric power cables transmit electrical energy over large distances using high-voltage, alternating current.

What are the advantages of using a high voltage and of using an alternating current?

	advantage of using a high voltage	advantage of using an alternating current
A	high current is produced in the cable	the resistance of the cable is reduced
B	high current is produced in the cable	the voltage can be changed using a transformer
C	less energy is wasted in the cable	the resistance of the cable is reduced
D	less energy is wasted in the cable	the voltage can be changed using a transformer

[ON2015/P11/Q35]

31 The diagram shows the magnetic field around wire X which carries a current into the paper.



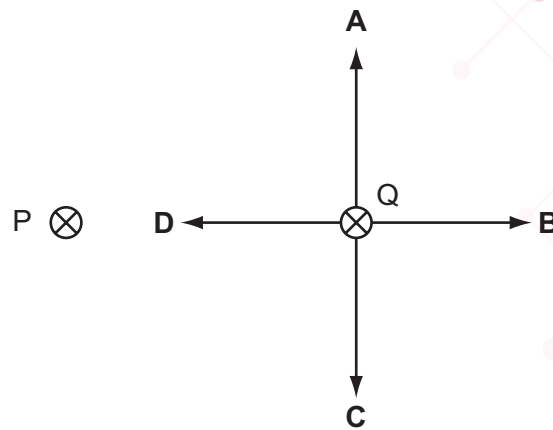
The arrows on the field lines show the direction of the force on

- A a N-pole.
- B a S-pole.
- C a small negative charge.
- D a small positive charge.

[ON2015/P12/Q31]

- 32 P and Q represent two, parallel, straight wires carrying currents into the plane of the paper. P and Q exert a force on each other.

Which arrow shows the force on Q?



[ON2015/P12/Q32]

P 1

Mark Scheme

1	B	21	C	41	C
2	C	22	D	42	C
3	B	23	B	43	C
4	B	24	D	44	D
5	A	25	C	45	B
6	A	26	D	46	B
7	D	27	D	47	D
8	A	28	C	48	A
9	B	29	A	49	B
10	C	30	A	50	B
11	A	31	A	51	D
12	D	32	A	52	A
13	B	33	C	53	D
14	C	34	B	54	D
15	B	35	D	55	B
16	C	36	D	56	D
17	D	37	A	57	A
18	B	38	D	58	D
19	B	39	B		
20	C	40	C		

- 01 (a) A wire carrying a current in a magnetic field experiences a force due to the current. On Fig. 1.1, insert the words **current**, **field** and **force** in the boxes to show the relative directions of the current, the magnetic field and the force.

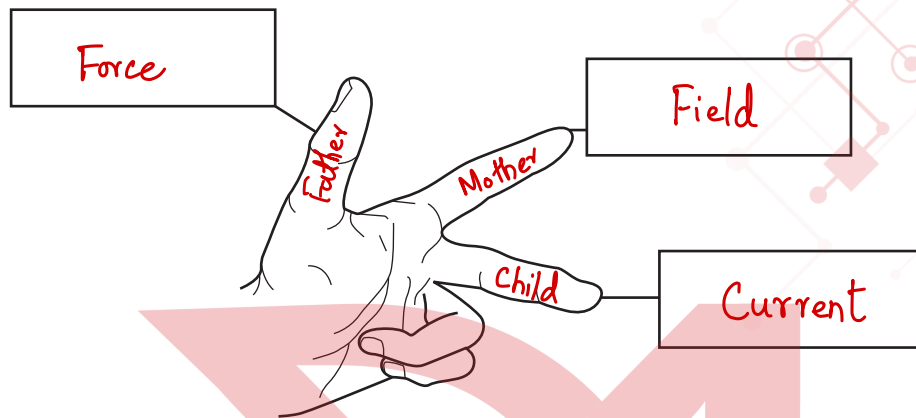


Fig. 1.1

[3]

[MJ2011/P21/Q8]

MOIZ DAWOODI
O/A LEVEL PHYSICS

02 Fig. 2.1 shows a rotating magnet in an alternating current generator that is used to power a lamp.

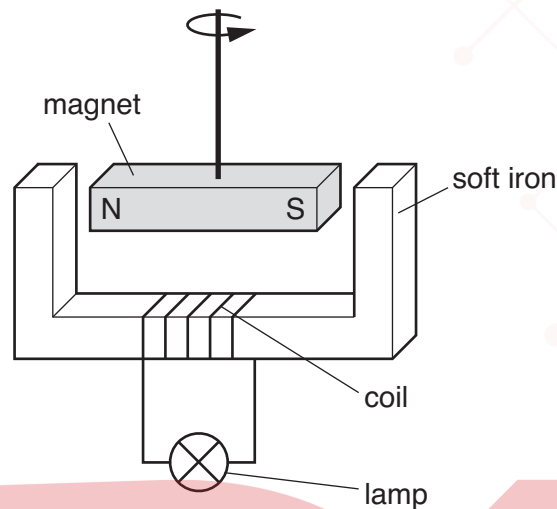


Fig. 2.1

- (a) (i) State how an alternating current differs from a direct current.

Magnitude of alternating current varies while magnitude Direct current remains constant.

[1]

- (ii) Explain, in detail, how alternating current is produced by the apparatus shown in Fig. 2.1.

As magnet rotates, field lines linking the coil changes, this changes flux, according to Faraday, emf will be induced. As the magnet moves, poles of magnet reverses after every half cycle so direction of induced current also reverses.

[4]

- (iii) State two ways in which the current in the lamp may be increased.

1. Increase no. of turns of coil or Use stronger magnet
2. Increase speed of rotation of coil.

[2]

(b) The generators at a power station produce a voltage of 25 000 V. This voltage is stepped up to 400 000 V by a transformer for long-distance transmission on overhead power lines. The voltage is later stepped down to 240 V.

(i) State and explain why the voltage is stepped up for long-distance transmission.

.....
.....
..... [2]

(ii) Calculate the ratio of the number of turns in the primary coil of the step-up transformer to the number of turns in its secondary coil.

ratio = [1]

(iii) State one advantage and one disadvantage of using thicker wire in the overhead power lines.

advantage:

.....

disadvantage:

..... [2]

(iv) An electric drill of power 1000 W is used in a country where the mains voltage is 240 V. State and explain the most appropriate fuse to use with this drill. You should select a fuse from the following values: 1 A, 3 A, 4 A, 13 A.

.....
.....
.....
..... [3]

[MJ2011/P21/Q9]

03 When a car is moving, its electrical equipment is powered by an a.c. generator.

(a) The coil of the a.c. generator is rotated by the car engine.

(i) On the axes in Fig. 3.1, sketch a graph of the output voltage of the coil against time for two rotations of the coil of the generator.

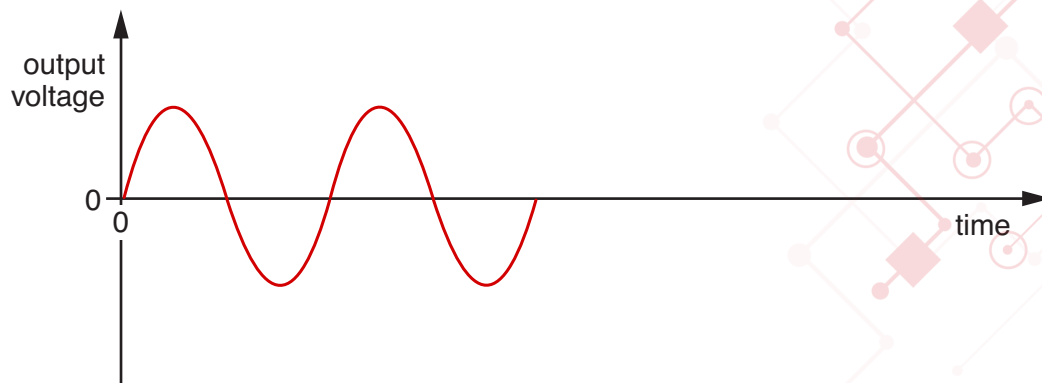


Fig. 3.1

[1]

(ii) The speed of the car increases and so does the speed of rotation of the coil.

State two changes that this causes to the output voltage.

change 1 Frequency will be increased

change 2 Emf will be increased.

[2]

[ON2011/P21/Q6]

MOIZ DAWOODI
O / A LEVEL PHYSICS

04 Fig. 4.1 shows part of an electric bell.

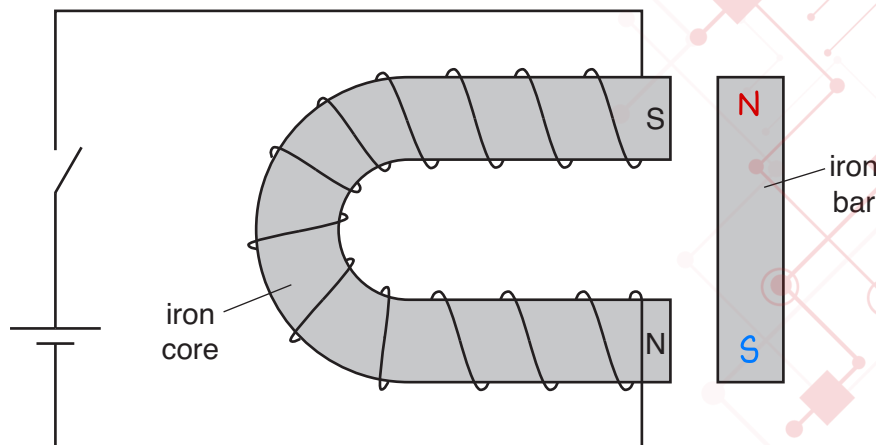


Fig. 4.1

A switch and a cell are in series with a length of wire coiled around an iron core.

The switch is closed and the current in the wire produces a south pole S and a north pole N at the ends of the core, as shown in Fig. 4.1. Magnetic poles are also produced in a small iron bar, placed near to the ends of the core.

- (a) (i) On Fig. 4.1, mark with an N the position of the north pole produced in the iron bar and mark with an S the position of the south pole produced in the iron bar. [1]

- (ii) State and explain what happens to the iron bar once it is magnetised.

Iron bar moves towards electromagnet as unlike poles attract.

.....[2]

- (b) The switch is opened and there is no current in the wire. State what happens to the magnetic poles in the iron bar.

No attraction will occur as iron will lose its poles.

.....[1]

[ON2011/P22/Q5]

- 05** Fig. 5.1 shows two coils of insulated wire wound on an iron ring. Coil A is connected to a battery and a switch. The switch is open. Coil B is connected to a sensitive centre-zero voltmeter.

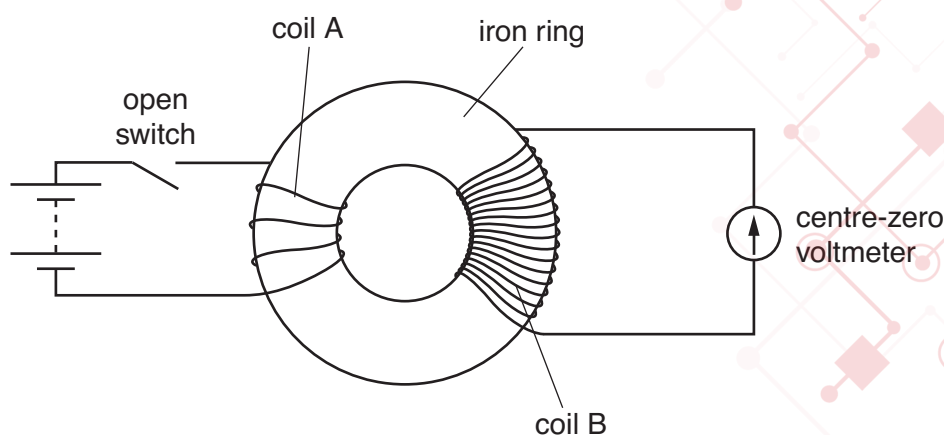


Fig. 5.1

The switch is closed. There is a current in coil A.

(a) On Fig. 5.1,

(i) mark the direction of the current in coil A, [1]

(ii) draw the magnetic field lines produced in the iron ring. [3]

(b) As the switch is closed, the voltmeter deflects to the right and then returns to zero.

(i) Explain why there is a deflection on the voltmeter.

.....

 [2]

(ii) The switch is opened. State and explain what happens to the deflection on the voltmeter.

.....

 [2]

(iii) Without changing coil A, state two changes to the apparatus that cause a greater deflection of the voltmeter.

1.
 2. [2]

[MJ2012/P22/Q9]

- 06 (a) Fig. 6.1 shows a solenoid made from wire wound around a plastic cylinder.

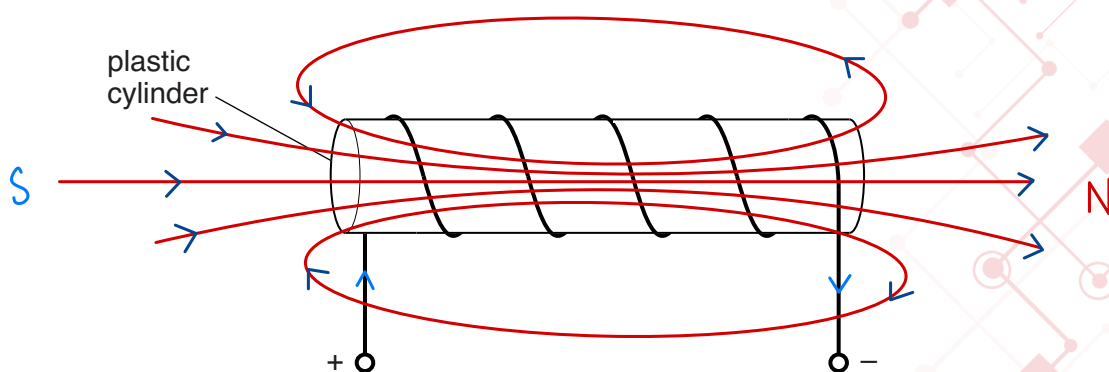


Fig. 6.1

A current in the solenoid produces a magnetic field.

On Fig. 6.1, draw the pattern of the magnetic field lines inside and outside the cylinder.

[2]

- (b) Fig. 6.2 shows a beam of beta-particles, in a vacuum, passing into a magnetic field.



Fig. 6.2

The movement of the beta-particles from left to right is an electric current.

- (i) On Fig. 6.2, draw an arrow to show the direction of the conventional current. [1]
- (ii) A solenoid is used to produce the magnetic field that lies within the shaded region of Fig. 6.2. The direction of the field is out of the page.

1. On Fig. 6.2, draw the path followed by one of the beta-particles in the shaded region. [2]
2. The direction of the current in the solenoid is reversed. State what happens to the path of the beta-particle.

.....
 [1]

[ON2012/P21/Q7]

07 Fig. 7.1 shows a simple transformer.

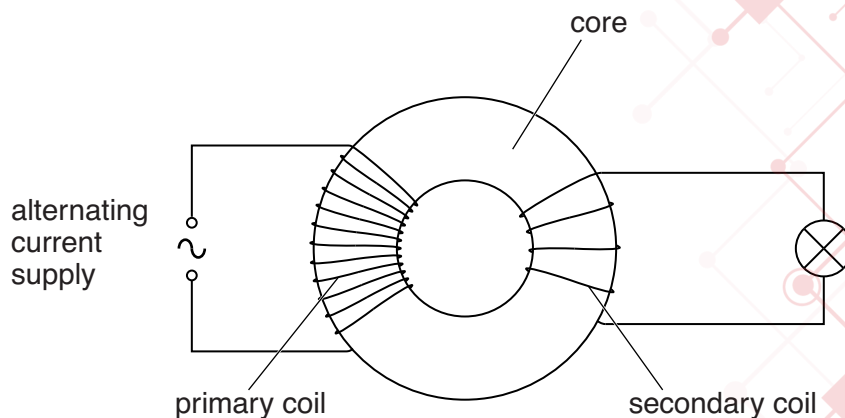


Fig. 7.1

(a) State the metal used for the core of a transformer.

..... [1]

(b) Explain how an alternating current in the primary coil causes the lamp to light.

.....
.....
.....
.....
.....
..... [3]

(c) Transformers are used to produce high voltages for the transmission of electrical power over long distances.

State one advantage of high voltage transmission.

.....
.....
..... [1]

[MJ2013/P21/Q8]

- 08** Fig. 8.1 shows a view, from above, of two wires X and Y. These wires carry equal currents vertically downwards through a piece of card.

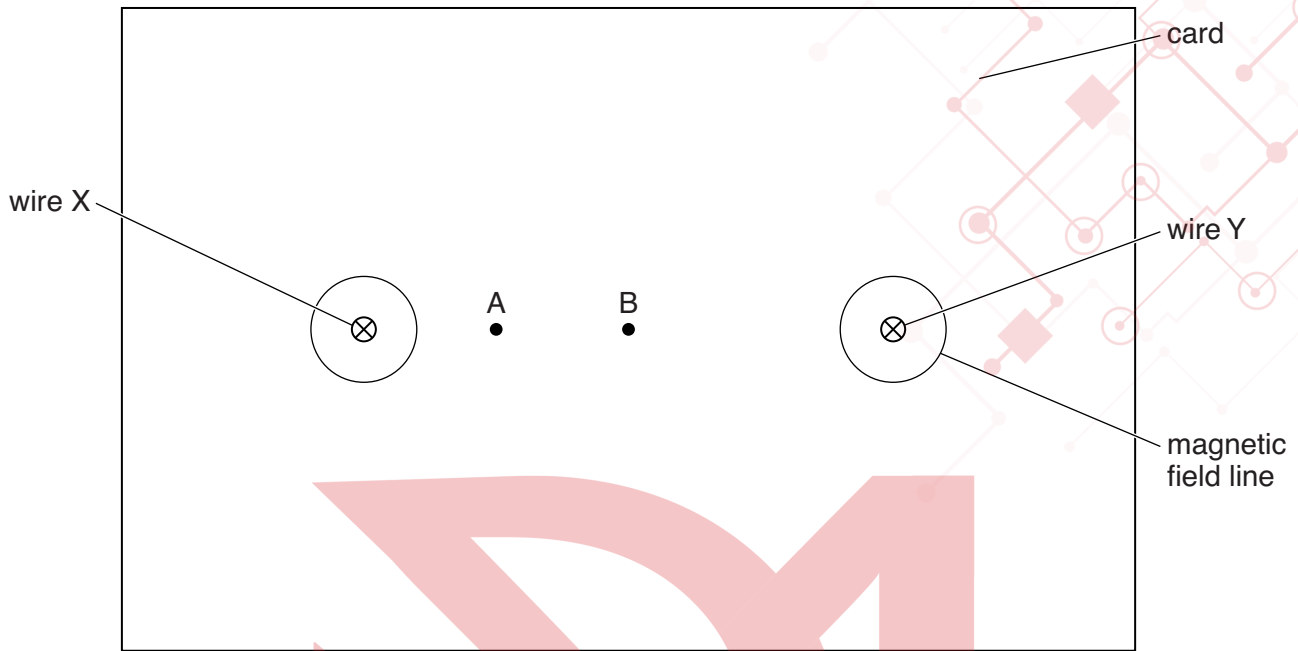


Fig. 8.1

One complete magnetic field line is drawn around each wire.

In this question, ignore the effects of the Earth's magnetic field.

- (a)** On Fig. 8.1, draw the complete magnetic field line due to the current in wire X that passes through point A. Mark the direction of this field line. [2]

- (b)** Point B is midway between the two wires. Explain why the magnetic field at B is zero.

.....
O/A LEVEL PHYSICS
..... [1]

- (c)** There is a force on wire Y due to the current in wire X.

- (i)** State the direction of the force on wire Y.

..... [1]

- (ii)** Explain why there is a force on wire Y.

.....
..... [1]

[MJ2013/P22/Q7]

9 EITHER

Fig. 9.1 shows a simple a.c. generator. The coil is turning and an e.m.f. is induced in the coil.

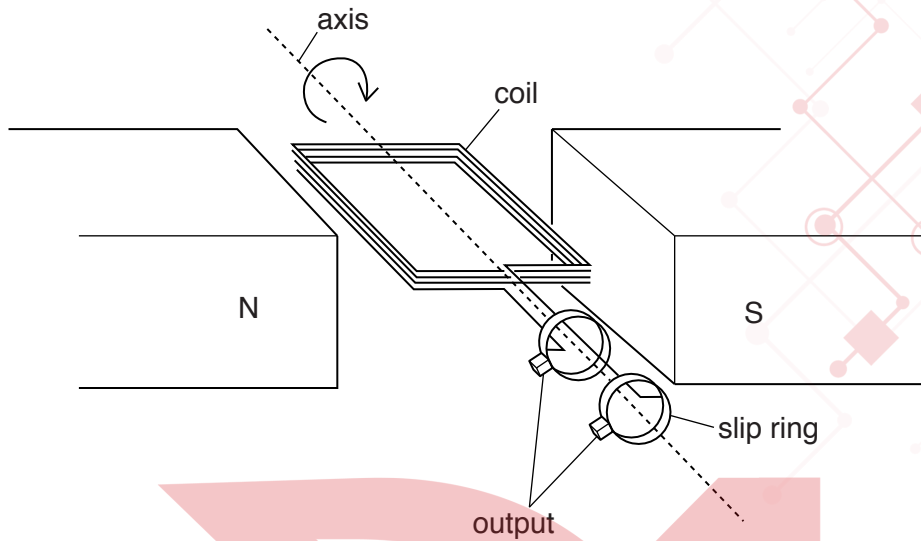


Fig. 9.1

- (a) The generator contains a permanent magnet. State the name of a metal used in a permanent magnet.

Steel

[1]

- (b) At the instant shown in Fig. 9.1, the induced e.m.f. is a maximum.

- (i) Explain why the induced e.m.f. is a maximum.

As no. of magnetic field lines linking the coil changed to maximum, change in flux becomes maximum. According to Faraday's law maximum emf will be induced in this case.

[2]

- (ii) State the position of the coil where there is no induced e.m.f.

Vertical position.

[1]

[MJ2013/P22/Q8]

10 Fig. 10.1 shows a coil of wire connected by flexible leads to a switch and a battery.

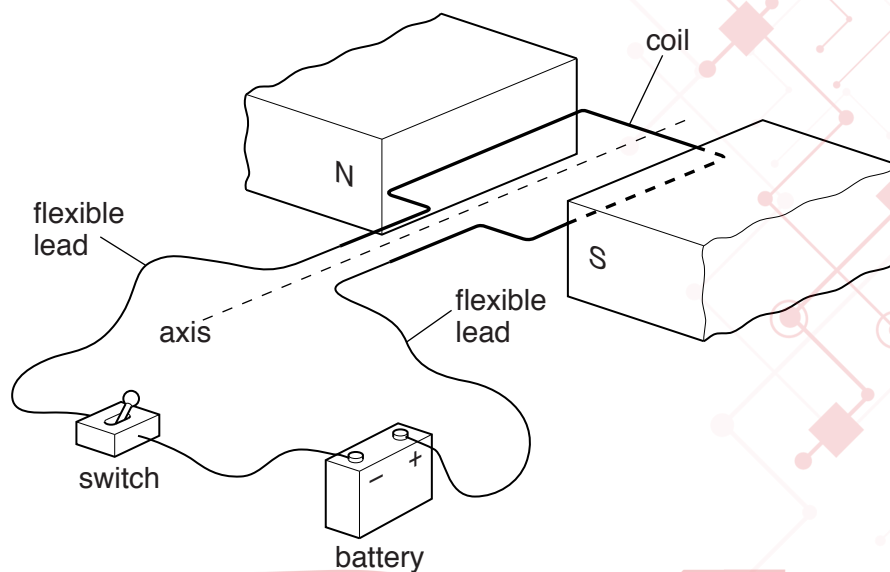


Fig. 10.1

The coil is placed between the poles of a permanent magnet and is free to turn about the axis.

When the switch is closed, forces due to the current act on the sides of the coil. The coil starts to turn.

(a) On Fig. 10.1, draw arrows to show the directions of the forces. [2]

(b) The coil stops when it is vertical. Explain why the turning effect of the forces is zero at this position.

..... [1]

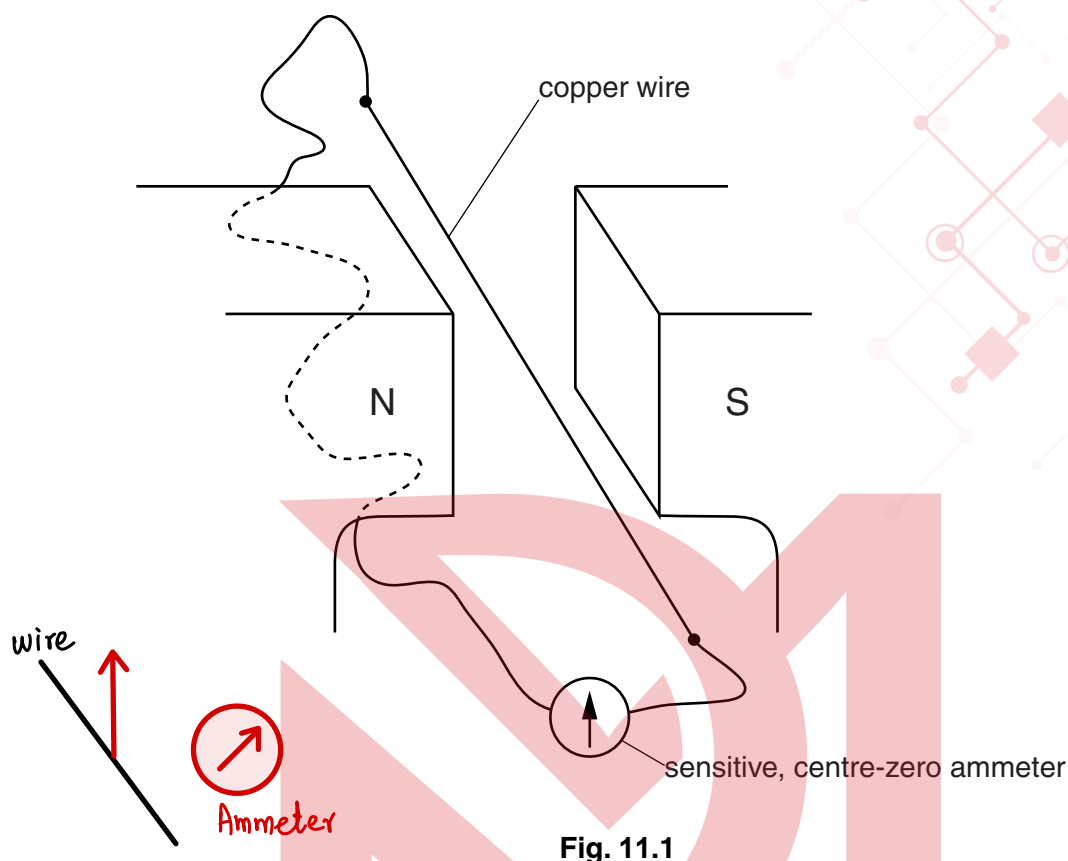
(c) In order for the coil to rotate continuously, a split-ring commutator is connected between the battery and the coil.

Explain how the split-ring commutator enables the coil to rotate continuously. Include a diagram in your answer.

..... [4]

[MJ2014/P21/Q6]

- 11 A straight length of copper wire lies horizontally between the poles of a U-shaped magnet. Fig. 11.1 shows the two ends of the wire connected to a very sensitive, centre-zero ammeter.



The copper wire is moved upwards slowly between the two magnetic poles. The needle on the ammeter deflects to the right.

- (a) Explain why the needle on the ammeter deflects.

As wire moves between poles, field lines linking the coil change, this changes flux. According to Faraday's Law, emf is induced $\mathcal{E}$, current flows so ammeter needle deflects. [2]

- (b) The wire is moved downwards very quickly between the two magnetic poles.

State what happens to the needle on the ammeter.

It will move to the left rapidly with greater deflection [1]

- (c) State what happens to the needle on the ammeter when the copper wire is moved horizontally between the two poles.

Remains at centered zero position. [1]

[ON2014/P21/Q7]

12 Fig. 12.1 shows the structure of a simple alternating current (a.c.) generator.

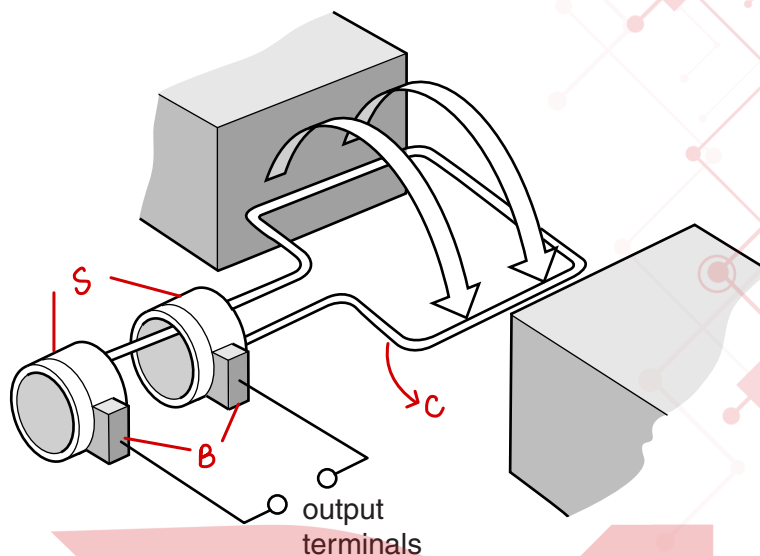


Fig. 12.1

(a) On Fig. 12.1, label

- the coil of the generator with the letter C,
- a slip ring with the letter S,
- a carbon brush with the letter B.

[2]

(b) The a.c. generator is operating and the arrows on Fig. 12.1 show the direction of rotation.

Explain why there is an electromotive force (e.m.f.) between the two output terminals.

As the coil moves, field lines linking the coil change, this changes flux. According to Faraday's Law, emf induces.

O/A LEVEL PHYSICS

[3]

[ON2014/P22/Q8]

13 A simple apparatus used to demonstrate electromagnetic induction is shown in Fig. 13.1.

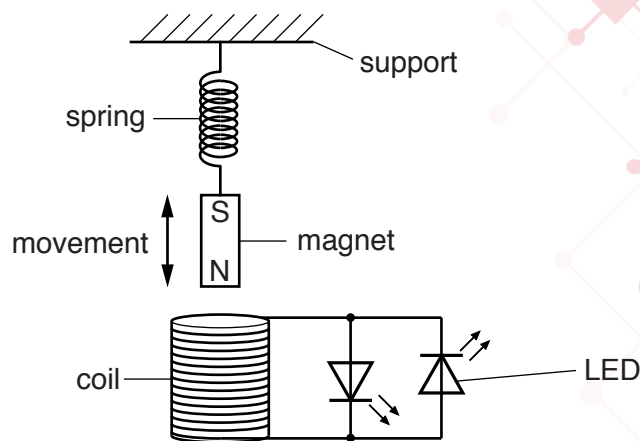


Fig. 13.1

The coil is connected to two light-emitting diodes (LEDs). The magnet moves into and out of the coil.

- (a) Explain why there is an induced e.m.f. in the coil when the magnet moves.

As the magnet moves, field lines linking the coil change, this changes flux. According to Faraday's Law, emf induces.

[2]

- (b) Explain why one LED lights up when the magnet moves into the coil and the other LED lights up when the magnet moves out of the coil.

As the magnet moves up, it makes one LED as forward biased & it glows while other is reversed biased, remain OFF. As the magnet moves down, the next LED glows & first is OFF.

[2]

- (c) The LEDs are brighter when the magnet moves faster. Explain why.

More change in flux causes more emf

[1]

[MJ2015/P21/Q7]

14 Fig. 14.1 shows a simple a.c. generator.

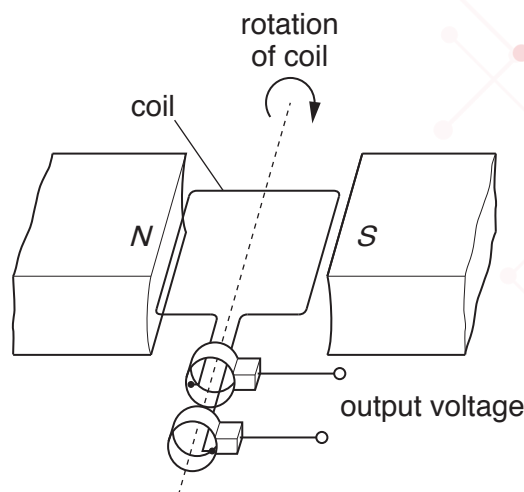


Fig. 14.1

- (a) The coil rotates and an alternating electromotive force (e.m.f.) is induced in the coil.

Fig. 14.2 shows how the alternating e.m.f. varies with time as the coil rotates.

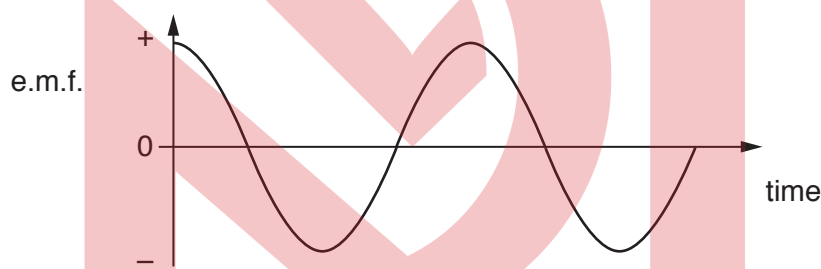


Fig. 14.2

Explain

- (i) why an e.m.f. is induced,

As the coil moves, field lines linking the coil change, this changes flux. According to Faraday's Law, emf induces.

[2]

- (ii) why the e.m.f. is sometimes positive and sometimes negative.

As coil rotates, it reverse its sides around poles of magnet hence direction of current & emf reverses.

[1]

(b) Changes are made to the a.c. generator, one at a time:

- stronger magnets are used
- more turns are wound on the coil
- the coil is turned faster.

Complete the table in Fig. 14.3 to show what happens to the maximum value of the e.m.f. and to the frequency of the alternating e.m.f.

changes made	what happens to the maximum value of the e.m.f.	what happens to the frequency of the e.m.f.
stronger magnets	increases	constant
more turns on the coil	increases	constant
the coil is turned faster	increases	increases

Fig. 14.3

[3]

[MJ2015/P22/Q6]

MOIZ DAWOODI
O / A LEVEL PHYSICS

- 15 (a) Fig. 15.1 shows a solenoid (long coil) X connected in series with a battery, a switch S and a variable resistor (rheostat).

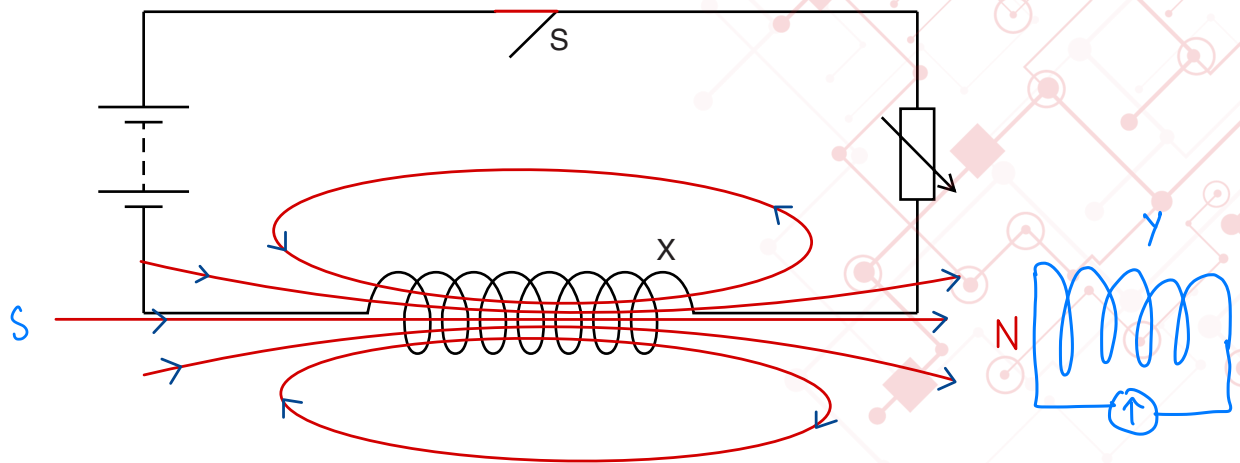


Fig. 15.1

The switch S is closed and there is a magnetic field due to the current in the solenoid.

- (i) On Fig. 15.1, draw the pattern of the magnetic field in, above and below the solenoid. [3]
- (ii) A second solenoid Y is connected to a sensitive centre-zero ammeter. Solenoid Y is placed in the magnetic field of X.
 1. The resistance of the variable resistor is gradually decreased and the pointer of the sensitive ammeter deflects slightly to one side.

Explain why this happens.

When resistance is decreased, current increases, this increases magnetic strength of solenoid hence change in magnetic flux occurs so emf induces & current flows.

[3]

2. The switch S is now opened.

Describe and explain what happens to the deflection on the ammeter as the switch is opened.

Current decreases to zero & solenoid X loses magnetism so now change in flux is negative. Needle deflects momentarily to opposite way & then returns to centered zero.

[3]

- (b) Fig. 15.2 shows a transformer that consists of two coils wound on an iron core. The transformer is connected to an electricity transmission cable.

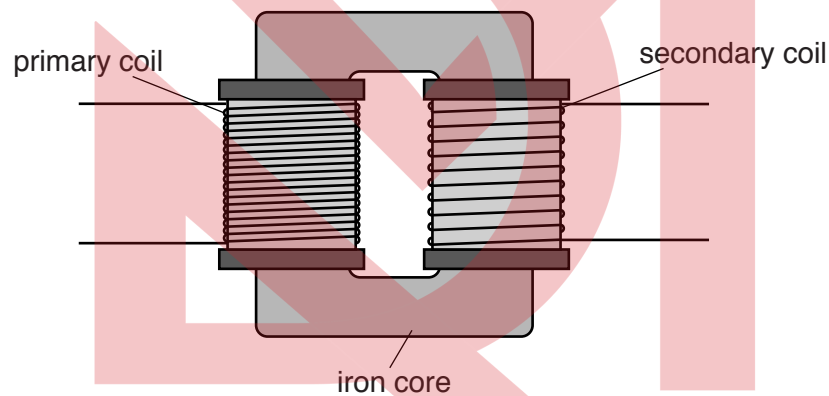


Fig. 15.2

- (i) Explain the purpose of the iron core in the transformer.

[2]

[ON2015/P21/Q10]

- 16 Thin wire, covered in plastic insulation, is used to make a solenoid (long coil). The solenoid is connected to a sensitive ammeter. Fig. 16.1 shows the N-pole of a steel magnet placed next to the solenoid.

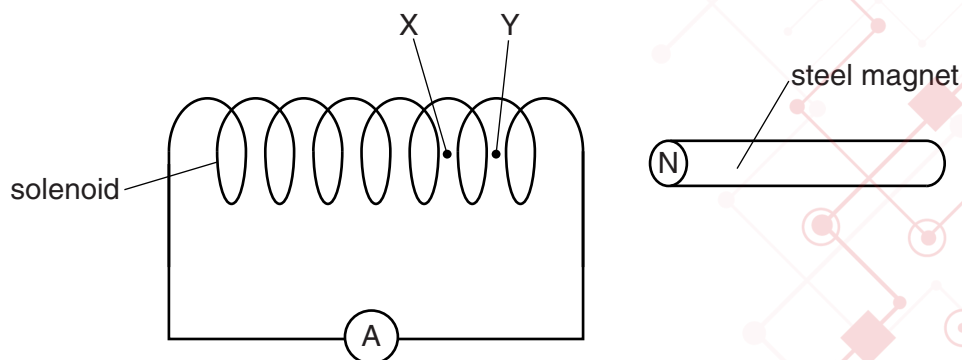


Fig. 16.1

Point X and point Y are on the axis of the solenoid.

- (a) (i) Explain why plastic is an electrical insulator.

Plastic doesn't contain free electron which are needed for conduction.

[1]

- (ii) Explain why the magnet is not made from

1. aluminium,

Non-magnetic material

[1]

2. iron.

Temporary magnetic material.

[1]

- (b) In one experiment, the magnet in Fig. 16.1 is moved to the left and passes into the solenoid. The N-pole of the magnet travels from Y to X at a constant speed. As it moves, the ammeter shows a small current.

- (i) Explain why there is a current in the solenoid when the magnet is moving.

When magnet moves, field lines linking the coil changes, flux changes,
According to Faraday's Law, emf induces so current flows.

[3]

[ON2015/P22/Q10]

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- 8 (a) all three correct: force, field, current B1

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- 9 (a) (i) d.c. current flows in one direction **or**
a.c. current flows in one direction then the other B1
- (ii) mention of magnetic field/flux B1
(magnetic) field lines // flux cuts coil // flux changes in coil B1
induction of voltage/current B1
something **relevant** reverses (e.g. field/flux cuts in one direction then the other // N pole approaches then leaves // N pole approaches and S pole approaches) B1
and link to a.c. B1
- (iii) **two** of: thicker wires; more turns of coil; stronger magnet; faster rotation; lower resistance (of lamp) B2
- (b) (i) to reduce heat/energy/power loss (on the power lines) B1
(higher voltage means) lower current B1
- (ii) 25:400 // 1:16 // 0.0625 B1
- (iii) reduces resistance // less power loss // costs less to run // more current // more power B1
increases weight // more support needed // more wind resistance // more ice forms // costs more to install B1
- (iv) 13 A B1
($I = P/V$) // 4.2 A // 4.17 etc A B1
must choose higher value to avoid fuse blowing // other fuses melt B1 [15]

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- 6 (a) (i) recognisable sine/cosine curve (≥ 2.0 cycles) B1
- (ii) larger (peak)(voltage) B1
higher frequency/shorter period/described in words (**allow** shorter wavelength) B1

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- 5 (a) (i) N at top end of bar **and** S at bottom end B1
- (ii) attracted to/moves towards iron core B1
unlike poles attract B1
- (b) they disappear/bar is demagnetised/loses its poles/is weaker B1 [4]

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- 9 (a) (i) conventional current direction correct in coil/one lead B1
- (ii) at least 1 line axially through coil A
OR line above and below end of coil A B1
- at least two curved lines in ring from ends of A
to ends of B (and inside A and B) B1
- correct direction on at least one line/arrow for candidate's (i) B1
- (b) (i) (magnetic) flux/field **cuts** (coil B)
OR field/flux changes (in coil B) B1
- induces** an e.m.f./voltage/current (in B) B1
- (ii) (voltmeter) deflects to left/opposite (and returns to zero) B1
- flux/field decreases/collapses/reduces
OR iron loses magnetism
OR **change** in field is in opposite direction
OR to oppose flux/field change B1
- (iii) ANY 2 lines
more turns on coil **B**
larger voltage/current (e.g. of battery)/more cells
battery smaller (internal) resistance
smaller resistance of wires; thicker wires; shorter wires
thicker or shorter iron ring; use **soft** iron;
coil A and B closer (on ring);
more sensitive voltmeter; laminate the iron ring B2

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- 7 (a) (at least) **two** parallel horizontal lines within the cylinder B1
(at least) **two** correctly shaped lines outside the cylinder B1
- (b) (i) ← (right to left) **and** on diagram (somewhere) B1
- (ii) 1. path continuously curving in same direction
upwards (**ignore** lines outside the shaded area) M1
A1
2. (changes to) downwards (curve) **not** reverses/opposite direction B1 [6]

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- 8 (a) (soft) iron/mu-metal B1
- (b) magnetic field **or** flux **or** flux/magnetic lines mentioned
changing magnetic field **or** changing flux **or** flux lines cut coil
induced voltage/current/e.m.f. B1
B1
B1
- (c) less power/energy/heat loss (**allow** no power loss/to prevent power loss) **or**
more efficient **or** thinner wire can be used (**ign.** cheaper) B1 [5]

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- 7 (a) oval/circle through or near A centered on or near X
clockwise arrow on line(s) around X **and** none wrong B1
B1
- (b) fields (due to X and Y) cancel **or** X and Y fields equal and opposite B1
- (c) (i) to the left
or towards X/A/B B1
- (ii) current (in wire Y) and (magnetic) field (caused by other wire)
or two (magnetic) fields interact B1 [5]

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8 EITHER

- (a) steel / magnadur / alnico / magnetite B1
- (b) (i) mention of cutting (lines of) magnetic field / change in (magnetic) flux
great(est) rate of change
or fast(est) cutting
or other explanation involving time M1
A1
- (ii) vertical/upright
or turned through 90°
or normal to (magnetic) field B1

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- 6 (a) arrows on long sides in opposite vertical directions [B1]
downwards on right **and** upwards on left **or** correct rotation shown [B1]
- (b) no (horizontal) distance between forces **or** forces through axle/pivot/axis [B1]
- (c) two halves of split ring clear and clearly connected to each end of coil [B1]
contacts/brushes labelled or described **and** connected to battery [B1]
each side of split ring touches other terminal/brush **or** current reverses in coil [B1]
or changes terminals of connection to battery [B1]
forces reverse on **sides** of coil **or** forces always in same direction on **side**
nearest a pole [B1]
- [7]

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- 7 (a) wire cuts field lines B1
current/e.m.f./voltage **induced** B1
- (b) larger deflection **and** to the left/opposite direction B1
- (c) no deflection/current B1 [4]

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- 8 (a) one label correct and not contradicted C1
C, 1S and 1B all correct and clear and none contradicted A1
- (b) any **three** from:
magnetic field (between poles)
(coil/wire) cuts field/flux **or** field/flux cuts (coil/wire) **or** field/flux changes
(electromagnetic) induction
brushes rub against/in contact with rings B3

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- 7 (a) field lines **of magnet** mentioned or magnetic flux mentioned B1
field lines cut the coil or flux changes B1
- (b) reversed movement of magnet causes one of B1
 - reversal of (induced) emf
 - reversal of (induced) current
 - field lines cut/flux change in reverse direction
LED emits light when **current** passes in one direction B1
- (c) more current or more induced emf B1
and flux lines cut faster or faster change in flux

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- 6 (a) (i) mention of (magnetic) field / flux (of N and S-poles)
(coil/wire) cuts **magnetic** field / flux / lines
or magnetic flux in coil changes C1
A1
- (ii) (one side of) coil cuts one way and then the other
or (side) moves one way and then the other / returns B1
or flux increases and then decreases
- (b) increase in emf for both stronger magnets and more turns B1
no change / same frequency for both stronger magnets and more turns B1
increase and increase for turn the coil faster B1

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- 10 (a) (i) at least two straight parallel lines inside the coil B1
at least two (complete) lines one above the coil and one below the coil B1
third line in middle **and** evenly spaced **and** two closed loops B1
(any crossings max. 2/3)
- (ii) current (in X) increases B1
magnetic field becomes stronger / changes B1
current / e.m.f. / voltage **induced** in Y / electromagnetic induction B1
opposite deflection B1
larger deflection B1
magnetic field decreasing **or** quicker (rate of) change B1 [9]
- (b) (i) to increase the strength of the magnetic field B1
to direct / concentrate the magnetic field (into the secondary coil) B1
- (ii) $(P =)VI$ **or** $33\,000 \times 85$ C1
 $2.8 \times 10^6 \text{ W}$ **or** 2800 kW **or** 2.8 MW A1
- (iii) $(E =)VIt$ **or** $33\,000 \times 85 \times 3600$ **or** $2.8 \times 10^6 \times 3600$ C1
 $1.0 / 1.01 / 1.008 \times 10^{10} \text{ J}$ A1 [6]

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- 10 (a) (i) no free electrons (in plastic) **or** all electrons are bound / structural B1
- (ii) (aluminium) is not magnet(ic) **or** cannot be magnetised B1
(iron) is a temporary / soft magnetic material **or** is not a permanent magnet B1 [3]
- (b) (i) magnetic field / flux (mentioned) B1
(magnetic) field lines cut wire / solenoid / circuit **or** changing magnetic field / flux B1
voltage / e.m.f. **induced** B1
- (ii) $(V =)IR$ **or** 0.045×1.2 **or** 0.000045×1.2 C1
 $5.4 \times 10^{-5} \text{ V}$ **or** 0.054 mV A1
 $(Q =)It$ **or** 0.045×0.14 **or** 0.000045×0.14 C1
 $6.3 \times 10^{-6} \text{ C}$ **or** 0.0063 mC A1 [7]